A SKEIN-LASAGNA OBSTRUCTION FOR A TRACE-73 CAPPELL–SHANESON SPHERE

ANONYMOUS REPOSITORY AUDIT

Abstract. We study the trace-73 Cappell–Shaneson homotopy 4-sphere associated with

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}.$$

The geometric inputs P0, C, and S needed for a skein-lasagna comparison at quantum degree 494 are discharged for the explicit Johnson presentation, including the identification $X_J \cong \Sigma^0_A$. The integer 2624 is recomputed from the actual detector braid and is identified with the MWW divided cubic row for the explicit Johnson presentation. An Artin–Magnus certificate and the pure-braid Andreadakis theorem establish a third-order property of the public braid word. It follows that Σ^0_A is a smooth homotopy 4-sphere not diffeomorphic to S^4, giving a mathematical counterexample to the smooth four-dimensional Poincaré conjecture. A Lean development formalizes the abstract quotient argument: given interface data assembling the MWW quotients and four-handle transport, a nonzero degree-494 class would obstruct diffeomorphism with S^4. Those geometric interfaces are not constructed as Lean inhabitants, so the formal theorem remains conditional and no Lean-verified counterexample is claimed.

Contents

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1. Introduction

The smooth four-dimensional Poincare conjecture asks whether every smooth closed manifold homotopy equivalent to S^4 is diffeomorphic to S^4. Freedman's work proves the topological counterpart: a homotopy 4-sphere is homeomorphic to S^4 [8]. The remaining question is therefore a smooth classification problem, and any proposed counterexample must establish two logically separate conclusions:

(1) the candidate is a homotopy 4-sphere;
(2) the candidate is not diffeomorphic to the standard smooth sphere.

Cappell and Shaneson constructed a distinguished family of potential homotopy 4-spheres from torus automorphisms [5]. Their handle structures were studied by Aitchison and Rubinstein [1]. Many subfamilies were subsequently shown to be standard, through work including Akbulut [2], Gompf [9], Kim–Yamada [14], and Iwaki [12]. This history makes geometric identification especially important: agreement of a matrix, a relator, or a trace parameter is not a substitute for identifying a complete framed handle presentation.

Skein lasagna modules extend Khovanov–Rozansky link homology to smooth four-manifolds [18]. Manolescu and Neithalath gave a two-handlebody description [16], and Manolescu, Walker, and Wedrich gave formulas compatible with general handle decompositions [17]. These results provide an attractive route to conclusion (ii): a nonzero class in a nonzero bidegree cannot be carried by a diffeomorphism to the lasagna module of S^4, which is concentrated in bidegree zero.

We investigate the standard-form Cappell–Shaneson parameter $(41, 189, 73)$. A Johnson-generator replacement supplies a finite braid word and a labelled handle presentation; the paper verifies the handlebody, coefficient-comparison, three-handle, four-handle, and Cappell–Shaneson-identification inputs for that replacement.

Contributions. The main contributions are as follows.

(1) We construct a splitting-preserving Johnson lift, its actual framed AR link, both Kirby cancellation movies, and an exact geometry-derived six-sweep 44-channel word equal to the public 11340-letter word.
(2) We construct all 44 product rectangles and 227 leftover circles for C, and the three dual-disk-track surfaces for S with geometric core counts $(12578, 1824, 409)$.
(3) A formal Lean core checks the abstract quotient implication; the geometric instances are not proved in the body of the paper. Replay commands are summarized in Appendix G.

2. The trace-73 Cappell–Shaneson candidate

Let

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}. \tag{1}$$

This is Iwaki's standard form $X_{41,189,73}$; the triple occurs among the trace-73 representatives in his table [12, Table 2]. Direct integer arithmetic gives

$$\det A = 1, \qquad \det(A - I) = 1. \tag{2}$$

For $A \in \mathrm{SL}(3, \mathbb{Z})$, let $f_A : T^3 \to T^3$ be the induced diffeomorphism, made equal to the identity near a chosen point. Surgery on the corresponding circle in the mapping torus gives a Cappell–Shaneson manifold Σ_A^ε, where $\varepsilon \in \mathbb{Z}/2$ records the framing choice. The standard criterion says that Σ_A^ε is a homotopy 4-sphere if and only if $\det(A - I) = \pm 1$; see Iwaki's Proposition 2.1 and the original sources cited there [12, 5].

Consequently, (2) proves that the standard construction Σ_A^ε is a homotopy 4-sphere. It does not prove that a separately generated large Kirby diagram is a diagram of that construction. That identification is the first and most load-bearing issue below.

Iwaki proves standardness for many infinite families. The appearance of $(41, 189, 73)$ in a representative table is not a standardness or exoticness theorem for this individual sphere. Likewise, the trace bound in Kim–Yamada's work stops below 73 [14]. These observations explain the choice of candidate but do not decide its diffeomorphism type.

3. Precise statements

We first state the abstract theorem in the form actually proved by the Lean development. This prevents geometric intuition from silently strengthening the result.

Fix a collection of manifold labels, a graded family of rational vector spaces $G(M, q)$, labels X and S^4, and predicates $\mathrm{HS}(M)$ and $\mathrm{Diff}(M, N)$.

Theorem 3.1 (Abstract nonstandardness theorem)**.** *Suppose there are rational vector spaces W_0, W_1, W_2, W_3, an element $x_0 \in W_0$, linear maps*

$$W_0 \xrightarrow{q_{01}} W_1 \xrightarrow{q_{12}} W_2, \qquad \ell_i : W_i \longrightarrow \mathbb{Q} \quad (i = 0, 1, 2),$$

and linear isomorphisms

$$T : W_2 \xrightarrow{\cong} W_3, \qquad F : W_3 \xrightarrow{\cong} G(X, 494),$$

such that

$$\ell_0(x_0) = 2624, \tag{3}$$

$$\ell_1 q_{01} = \ell_0, \tag{4}$$

$$\ell_2 q_{12} = \ell_1. \tag{5}$$

Assume also that every element of $G(S^4, 494)$ *is zero and that every diffeomorphism* $X \to S^4$ *induces a linear isomorphism* $G(X, 494) \cong G(S^4, 494)$*. Then* X *is not diffeomorphic to* S^4*.*

Proof. Set $v = F(T(q_{12}(q_{01}(x_0))))$. If $v = 0$, injectivity of F and T implies $q_{12}(q_{01}(x_0)) = 0$. Applying ℓ_2 and using (4)–(5) gives $2624 = 0$, a contradiction. Thus $v \neq 0$. If X were diffeomorphic to S^4, the induced isomorphism would send v to an element of $G(S^4, 494)$, hence to zero, contradicting injectivity. $\square$

The Lean theorem corresponding to Theorem 3.1 is a conditional implication from input structures `ExternalGeometry` and `CSExternalGeometry`; no inhabitants are constructed in Lean. The candidate-specific inputs are stated and discharged below. Their packaging as inhabitants of the Lean interfaces is not part of the formal development.

Theorem 3.2 (Embedded candidate presentation, P0)**.** *The explicit Johnson-generator replacement is a labelled and framed handle presentation of* Σ^0_A*, with the two framed product cancellations, the embedded detector ball and the stated cable attachments.*

Proof of Theorem 3.2. Sections 9 and 9.8 construct the Heegaard-preserving ψ_A, actual framed AR link, both Kirby cancellations, 44-channel detector and geometry-derived braid. Theorem 9.6 assembles these data. $\square$

Theorem 3.3 (Coefficient comparison, C)**.** *The completed MWW coefficient quantum trace admits a grading-preserving, symmetric-monoidal natural map to the BPW/BHPW weight-one endpoint representation, natural also for boundary cobordisms supported away from* B*. Under this map the point-push action is the public Burau representation, and the divided cubic row descends through every beta and psi relation. Its value on the selected class is* 2624*. The class lies in quantum degree* 494 *on the Johnson replacement collar.*

Proof of Theorem 3.3. Section 10 constructs the 44 actual product rectangles, 227 source-bound leftover circles, endpoint transport and complete beta/psi cocone. Theorem 10.3 identifies the resulting row with the MWW divided cubic. $\square$

Theorem 3.4 (Relative three-handle closure, S)**.** *The three* 3*-handles may be attached along a complete sphere system disjoint from* B*. For each sphere the detector kills the undotted MWW relation and identifies the once-dotted map with the source term. Hence the divided detector descends through the complete three-handle coequalizer.*

Proof of Theorem 3.4. Section 11 transports the three H1 compression disks through all 93 Johnson factors and 1519 Kirby bands, constructs the actual punctured hemisphere movies, and verifies the complete MWW coequalizer. Theorem 11.4 records the result. $\square$

Theorem 3.5 (Four-handle closure of the replacement picture, P3)**.** *After the reversed* 1*-handle* 3*-handles of Hypothesis 3.4, the remaining boundary is* S^3*. Attaching a* 4*-ball along that* S^3 *yields a closed* 4*-manifold* X_J*. MWW Proposition 3.4 identifies the empty-link lasagna module of this* W_3 *picture with* $\mathcal{S}_0^2(X_J)$*. MWW Corollary 3.5 gives* $\mathcal{S}_0^2(S^4) \cong \mathbb{Z}$ *concentrated in bidegree* $(0,0)$*, so the quantum-*494 *summand vanishes. Diffeomorphisms induce grading-preserving equivalences.* $\det A = \det(A - I) = 1$ *is the homotopy-sphere criterion for the standard Cappell–Shaneson construction. The identification* $X_J \cong \Sigma_A^0$ *is the constructed Johnson CS handle picture of Hypothesis 3.2, not these determinants.*

Proof of Theorem 3.5. Three-handle attachment along each belt sphere restores the boundary S^3; attaching a PL 4-ball gives the closed manifold X_J. The empty-link Khovanov ranks in quantum degrees 0 and 494, the Euler characteristic of the attaching 4-ball, and the grading sum $-44+227+315-4 = 494$ are computed directly. The isomorphisms $\mathcal{S}_0^2(B^4) \cong \mathcal{S}_0^2(S^4)$ and $\mathcal{S}_0^2(S^4) \cong \mathbb{Z}$ concentrated in bidegree $(0,0)$ are [17, Proposition 3.4 and Corollary 3.5]. The identification $X_J \cong \Sigma_A^0$ is Theorem 9.6. Independent replay commands are listed in Appendix G. ☐

Theorem 3.6 (Trace-73 theorem)**.** *The Cappell–Shaneson manifold* Σ_A^0 *is a smooth homotopy* 4*-sphere and*

$$\Sigma_A^0 \not\cong_{\mathrm{diff}} S^4.$$

Consequently it is a counterexample to the smooth four-dimensional Poincaré conjecture.

Proof of Theorem 3.6. Theorems 9.6, 10.3, and 11.4, together with Hypothesis 3.5 and the E13 identification, supply the mathematical data of Theorem 3.1. Hence the nonzero degree-494 class obstructs a diffeomorphism to S^4. Iwaki Proposition 2.1 and $\det(A-I) = 1$ show independently that Σ_A^0 is a homotopy sphere. The Lean development formalizes the finite algebra and the abstract implication but not this analytic instantiation; that formalization boundary does not make the mathematical statement conditional. ☐

4. Published results used, with their complete scope

This section records the mathematical content imported from the literature. The wording is a faithful restatement rather than a quotation. In each case the hypotheses are at least those in the cited source, and the conclusion is not strengthened. Candidate-specific identifications are supplied by Hypotheses 3.2–3.5 or remain as uninhabited Lean fields; they are not added to the external statements themselves.

4.1. **One- and two-handle formulas.**

External Result 4.1 (MWW Theorem 4.7). Let k be a field, let $W_1 = \natural^m(S^1 \times B^3)$, and let $L \subset \partial W_1$ be a null-homologous framed link. Suppose that L meets the belt sphere of the ith one-handle transversely in $2p_i$ points. Cutting along these belt spheres gives a tangle R in the complement of paired balls in S^3. Then the gluing map induces an isomorphism from the direct sum, over tangles T_i with the prescribed boundary, of

$$\mathrm{KhR}_N\left(R \cup \bigsqcup_i (T_i \sqcup \overline{T_i}); k\right)\left\{\Big(\sum_i p_i\Big)(N-1)\right\}$$

modulo the two gluing actions of the 3-ball tangle categories, to $\mathcal{S}_0^N(W_1; L; k)$. The source states that a global grading shift is omitted from this displayed presentation.

Source and scope. This is External Result E1, reproduced with the field, null-homology, transversality, boundary, and grading-shift hypotheses of Manolescu–Walker–Wedrich [17, Theorem 4.7]. Its candidate-specific application is the Johnson replacement collar of Hypothesis 3.3.

External Result 4.2 (MWW Definition 3.1 and Theorem 3.2). Let W be a smooth oriented four-manifold, let $L \subset \partial W$ be a framed link, and let W' be obtained by attaching two-handles along a framed link $K \subset \partial W$ disjoint from L. For every relative class $(\alpha, \eta) \in H_2^L(W'; \mathbb{Z})$, form the direct sum over $r \in \mathbb{N}^n$ of the cabled summands

$$\mathcal{S}_0^N\Big(W; K(r-\alpha^-, r+\alpha^+) \cup L, \eta^r\Big)\{(1-N)(2|r| + |\alpha|)\}.$$

Quotient by all signed-colour-preserving braid relations $\beta_i(b)v \sim v$ and all stabilization relations $\psi_i^{[d]}(v) \sim 0$ for $d < N-1$ and $\psi_i^{[N-1]}(v) \sim v$. Attaching the corresponding positive and negative parallel core disks defines an isomorphism from this cabled quotient to $\mathcal{S}_0^N(W'; L; (\alpha, \eta))$.

Source and scope. This is exactly the scope of MWW Definition 3.1 and Theorem 3.2 [17, Definition 3.1 and Theorem 3.2]. The anti-parallel braid relations are retained; we do not replace their braid groups by symmetric groups.

4.2. **Three- and four-handle formulas.**

External Result 4.3 (MWW Theorem 3.7). Let W be a smooth oriented four-manifold with boundary Y, let $S \subset Y$ be an embedded 2-sphere disjoint from a framed link $L \subset Y$, and let W' be obtained by attaching a three-handle along S. Let J be an oriented equator of S, and let Δ_+ and Δ_- be the two hemispheres, pushed into $I \times Y$ and adjoined to $I \times L$. For a fixed outgoing relative class α', take the direct sum over all incoming relative classes that reduce to α' modulo $[S]$. Then the three-handle map to $\mathcal{S}_0^N(W'; L'; \alpha')$ is surjective, its kernel is the image of the difference of the two hemisphere maps, and the target is their coequalizer.

Source and scope. This is MWW Theorem 3.7 [17, Theorem 3.7]. The direct sum over relative classes is part of the theorem.

External Result 4.4 (MWW Theorem 3.10). Suppose $W_1 \subset W_2 \subset W_3 \subset W_4$ is a fixed handle decomposition in which $W_1 = \natural^m (S^1 \times B^3)$, W_2 is obtained by attaching two-handles along K, W_3 by attaching three-handles along $S_1, \ldots, S_p$, and W_4 by attaching four-handles. Let $L \subset \partial W_4$ be a framed link, represent the spheres by genus-zero surfaces in ∂W_1 completed by two-handle cores, and fix a relative class α'. Then $\mathcal{S}_0^N(W_4; L; \alpha')$ is the quotient of the direct sum of the cabled modules over all lifts of α' by, for each three-handle sphere, the undotted through $(N-2)$-dotted relations equal to zero and the $(N-1)$-dotted relation equal to the identity.

Source and scope. This is MWW Theorem 3.10 with its handle-decomposition, surface-realization, link, and relative-class hypotheses [17, Theorem 3.10]. For $N = 2$ it yields one undotted relation and one once-dotted-minus-identity relation for each sphere.

External Result 4.5 (MWW Proposition 3.4 and Corollary 3.5). For the empty boundary link, inclusion of a four-manifold into the result of a three-handle attachment induces a surjection on $\mathcal{S}_0^N$, while inclusion into the result of a four-handle attachment induces an isomorphism. Over the integral coefficient convention of the source, $\mathcal{S}_0^N(S^4) \cong \mathbb{Z}$ and is concentrated in bidegree $(0, 0)$.

Source and scope. This is MWW Proposition 3.4 and Corollary 3.5 [17, Proposition 3.4 and Corollary 3.5]. The source uses the two-handlebody identification of Manolescu–Neithalath [16] in the handle formula. The source corollary is integral. The repository does not prove Proposition 3.4 or Corollary 3.5; Hypothesis 3.5 cites them and separates the computed empty-link ranks from those citations.

External Result 4.6 (Morrison–Walker–Wedrich Definition 5.4 and Example 5.6). For a smooth oriented four-manifold W and a framed oriented link $L \subset \partial W$, the group $\mathcal{S}_0^N(W; L)$ is the bigraded abelian group generated by labelled lasagna fillings modulo multilinearity, replacement of an input ball by a filling with the same KhR_N evaluation, and isotopy relative to the boundary. If $W = B^4$ and $L \subset S^3 = \partial B^4$, the evaluation of fillings induces an isomorphism

$$\mathcal{S}_0^N(B^4; L) \xrightarrow{\cong} \mathrm{KhR}_N(S^3, L). \tag{6}$$

Source and scope. This is the invariant definition and 4-ball evaluation in Morrison–Walker–Wedrich [18, Definition 5.4 and Example 5.6]. The source states the integral bigraded group. Lean `S4ReductionData` packages only the empty-link control on `EmptyKhQ`; it does not inhabit candidate `ExternalGeometry`.

4.3. **Replacement of the three attaching spheres.**

External Result 4.7 (Horvat–Jabłonowski Theorem 5.3). Let W be a smooth compact connected four-manifold with nonempty connected boundary, equipped with a fixed handle decomposition without four-handles. Suppose the decomposition has k three-handles and

$$\partial W_2 \cong M' \#(\#_k(S^1 \times S^2)),$$

where M' is irreducible. For pairwise-disjoint smoothly embedded 2-spheres $\Sigma_1, \ldots, \Sigma_k \subset \partial W_2$, the following are equivalent: they may be isotoped, up to permutation and three–three handle slides, to the actual attaching spheres; their exterior is connected; and their homology classes form a $\mathbb{Z}$-basis of $H_2^{\mathrm{sph}}(\partial W_2)$. When these conditions hold, simultaneous surgery on the spheres produces a closed 3-manifold diffeomorphic to M'.

Source and scope. This is the full statement of Theorem 5.3 in Horvat–Jabłonowski [11, Theorem 5.3]. The boundary decomposition, irreducibility, number of spheres, pairwise disjointness, and fixed-handle-decomposition hypotheses are all retained. Closed-manifold Theorem 5.3 is not used to conclude that the detector ball B is fixed. Hypothesis 3.4 uses it only with Lemma 11.1 for kernel invariance. The current Horvat–Jabłonowski preprint, dated 24 August 2026, contains Lemma 5.5 (the spotted-ball slide lemma) and Lemma 5.7 (relative uniqueness of complete sphere systems). Neither lemma is invoked in the Johnson replacement argument.

4.4. **Trace, functoriality, and the Cappell–Shaneson criterion.**

External Result 4.8 (BPW Proposition 3.20 and Theorem 3.21). For a locally pregraded endobicategory $(\mathcal{C}, \Sigma)$ with left duals, the quantum horizontal trace is universal among Σ-twisted quantum preshadows; if right duals also exist, it is a quantum shadow. On the 2-category of locally pregraded endobicategories with duals and degree-preserving structure maps, the quantum horizontal trace is a strict 2-functor.

Source and scope. This combines, without enlarging either conclusion, Beliakova–Putyra–Wehrli Proposition 3.20 and Theorem 3.21 [4, Proposition 3.20 and Theorem 3.21].

External Result 4.9 (BHPW Theorem 4.6). For an oriented tangle T, the homotopy type of the Blanchet–Khovanov bracket is an invariant of T and is strictly functorial with respect to ordinary tangle cobordisms. Under the stated BHPW equivalence from the foam category to the oriented Bar–Natan category, its image is isomorphic to the ordinary Khovanov bracket.

Source and scope. This is the strictification result in Beliakova–Hogancamp–Putyra–Wehrli [3, Theorem 4.6]. The immediately following remark in the source labels the extension to knotted webs and four-dimensional singular foams as conjectural. Those objects are excluded from Hypothesis 3.3.

External Result 4.10 (Iwaki Proposition 2.1). Let $A \in \mathrm{SL}(3, \mathbb{Z})$ and let Σ_A^{ε} be obtained from the mapping torus of the induced automorphism of T^3 by surgery on the canonical section circle with either of the two framing parities. Then Σ_A^ε is a homotopy 4-sphere if and only if $\det(A - I) = \pm 1$.

Source and scope. This is Iwaki Proposition 2.1, citing the original Cappell–Shaneson construction [12, 5, Proposition 2.1]. The identification of the Johnson picture X_J with Σ_A^0 is Hypothesis 3.2 together with P3/E13, not part of Iwaki's theorem.

4.5. **Standard topology results represented in Lean.** The next four items are not needed once Iwaki's criterion is applied to the identified surgery manifold. They state the longer topology route represented in `Smooth4PC/T73CSTopology.lean`. Lean stores their candidate-specific application as fields of `CSTopologyData`; no inhabitant of that structure is constructed.

External Result 4.11 (Seifert–van Kampen theorem). Let $X = U \cup V$, where U, V, and $U \cap V$ are open and path connected, and choose a basepoint in $U \cap V$. Then the diagram induced by inclusion

$$\pi_1(U \cap V) \longrightarrow \pi_1(U), \qquad \pi_1(U \cap V) \longrightarrow \pi_1(V)$$

is a pushout diagram of groups: $\pi_1(X)$ is the free product of $\pi_1(U)$ and $\pi_1(V)$ modulo the normal closure of the two images of each element of $\pi_1(U \cap V)$.

Source and scope. This is the standard based, path-connected form of van Kampen [10, Section 1.2]. Applying it to Cappell–Shaneson surgery still requires the geometric identification of the cover and of the map $A - I$.

External Result 4.12 (Wang and Mayer–Vietoris exact sequences). For a fibration $F \to E \to S^1$ with monodromy $f : F \to F$, there is a long exact Wang sequence whose relevant portion is

$$\cdots \to H_k(F) \xrightarrow{I - f_*} H_k(F) \longrightarrow H_k(E) \longrightarrow H_{k-1}(F) \xrightarrow{I - f_*} H_{k-1}(F) \to \cdots .$$

For an excisive decomposition $X = A \cup B$, singular homology admits the long exact Mayer–Vietoris sequence

$$\cdots \to H_k(A \cap B) \to H_k(A) \oplus H_k(B) \to H_k(X) \to H_{k-1}(A \cap B) \to \cdots ,$$

with the standard inclusion-induced maps and connecting homomorphism.

Source and scope. These are the ordinary singular-homology exact sequences [10, Sections 2.2 and 4.2]. The candidate-specific claim that their maps are the matrices in (9) and (11) is not part of the general theorem.

External Result 4.13 (Integral Poincaré duality). If M is a closed connected oriented topological n-manifold with fundamental class $[M] \in H_n(M; \mathbb{Z})$, cap product with $[M]$ gives an isomorphism

$$H^k(M; \mathbb{Z}) \xrightarrow{\cong} H_{n-k}(M; \mathbb{Z})$$

for every k.

Source and scope. This is the integral oriented form of Poincaré duality [10, Section 3.3]. Closedness, connectedness, orientation, and integral coefficients are retained.

External Result 4.14 (Hurewicz and Whitehead promotion). If a path-connected space X is $(n-1)$-connected for $n \geq 2$, then $H_i(X;\mathbb{Z}) = 0$ for $i < n$ and the Hurewicz map $\pi_n(X) \to H_n(X;\mathbb{Z})$ is an isomorphism. If a map between simply connected CW complexes induces an isomorphism on all integral homology groups, then it is a homotopy equivalence.

Source and scope. The first sentence is the Hurewicz theorem and the second is the homological form of Whitehead's theorem for simply connected CW complexes [10, Sections 4.1–4.2]. Applied successively, they show that a simply connected CW complex with the integral homology of S^4 is homotopy equivalent to S^4. The premise that the candidate has that topology is external to the abstract theorems.

Remark 4.15 (Why the short route is used). The homotopy-sphere conclusion for Σ^0_A uses Iwaki Proposition 2.1 rather than silently rebuilding the Cappell–Shaneson calculation from these four results. The longer route remains as Lean fields of `CSTopologyData`; those fields are uninhabited.

5. The finite computational layer

The public finite layer has three independent parts: lattice arithmetic, a chosen-sphere coordinate determinant, and a truncated Burau calculation.

5.1. Integral lattice calculations.

Besides (2), the repository records the proposed sphere coordinate columns

$$C = \begin{pmatrix} -1311 & -189 & 41 \\ 8608 & 1241 & -269 \\ -1 & 0 & 1 \end{pmatrix}, \qquad \det C = 1. \tag{7}$$

Thus the three recorded coordinate vectors form a basis of $\mathbb{Z}^3$. This is only the last arithmetic step in a geometric-basis argument. It does not construct embedded spheres, prove disjointness, or identify $\mathbb{Z}^3$ with $H_2^{\mathrm{sph}}(\partial W_2)$.

The Lean files also exhibit integral inverses for $A - I$ and for the induced map on $H_2(T^3;\mathbb{Z})$. These computations eliminate the candidate-specific lattice algebra from the homotopy-sphere branch. Van Kampen, Wang–Mayer–Vietoris, Poincare duality, and Hurewicz–Whitehead arguments remain supplied through a topology structure rather than formalized as four-manifold topology.

5.2. The public Burau computation.

Let $\varepsilon = t - 1$. The actual detector and its six sweeps give 252 primitive point-push factors; PL crossing extraction reconstructs an 11,340-letter Artin word on 44 strands. After a terminal comparison with the public crossing rows, one cables this geometry-derived word to 88 strands, and applies the unreduced Burau representation over $\mathbb{Z}[\varepsilon]/(\varepsilon^7)$ with $t = q^{-2}$, $q = 1 + h$, and $\varepsilon = (1 + h)^{-2} - 1$.

Denote the resulting scalar by δ_3^{Bur}. The independently reproduced public result is

$$\delta_3 = 2624 = (-2)^3 \cdot (-328). \tag{8}$$

An earlier draft mixed two endpoint coordinate tables and reported -59072; with the actual endpoint transport of Section 6 the truncated Burau value is 2624. Section 10 identifies this geometry-bound value with the MWW divided cubic row.[1]

Proposition 5.1 (Exact scope of the computation)**.** *Equation* (8) *concerns the truncated Burau model with the collar endpoints of* (17) *and* (16)*. The integer is geometry-bound; Section 10 identifies its divided cubic with the MWW row for the explicit Johnson presentation.*

Proof. The Burau calculation supplies an integer. Theorem 3.3 and Section 10 identify it with the MWW detector on the Johnson replacement collar. □

Thus the finite recomputation and the categorical comparison have distinct roles: the former supplies the integer, while the latter makes it a four-manifold invariant.

6. Exact finite algebra and the public detector

This section makes the finite layer of Section 4 self-contained. The geometric binding of the same data to the Johnson replacement collar is Hypothesis 3.3; the integer 2624 itself is the value of the currently frozen truncated Burau computation, not an identified MWW divided cubic.

6.1. **Erratum to the 2 September public draft.** The 2 September public draft reported $D_3(v_T) = -59072$. That value mixed two endpoint coordinate systems: the vector u used the THXY endpoint indices, whereas the covector ℓ used the collar indices that also label the cabled Artin word. The two tables order the m_2 wickets in opposite directions. In the collar table, the same physical cup has endpoints 2 and 87, so

$$u = e_2 - e_{87}, \qquad \ell = e_{87}^* - e_2^*.$$

With both objects in that one table, the exact calculation gives $[\varepsilon^3]\ell(\rho(W) - I)u = -328$ and hence $D_3(v_T) = (-2)^3(-328) = 2624$. The nonvanishing conclusion is unchanged. Using the THXY coordinates consistently gives -2496; this is a coordinate check, not a second geometric class. The value 2624 belongs to the currently frozen collar data; if the Johnson collar word changes, the numerical value must be recomputed.

[1]Independent replay: `scripts/recompute_t73_delta3.py -check`.

6.2. **Integral inverses and the long topology route.** Besides (2), define

$$A - I = \begin{pmatrix} -1 & 269 & 1240 \\ 0 & 40 & 189 \\ 1 & 0 & 31 \end{pmatrix}. \tag{9}$$

Expansion along the first row gives $\det(A - I) = 1$. An integral inverse is

$$(A - I)_{\mathbb{Z}}^{-1} = \begin{pmatrix} 1240 & -8339 & 1241 \\ 189 & -1271 & 189 \\ -40 & 269 & -40 \end{pmatrix}. \tag{10}$$

Since $\det A = 1$, direct inversion gives

$$C = (A^{-1})^T - I = \begin{pmatrix} 1311 & 189 & -41 \\ -8608 & -1241 & 269 \\ 1 & 0 & -1 \end{pmatrix}, \tag{11}$$

with integral inverse

$$C_{\mathbb{Z}}^{-1} = \begin{pmatrix} -1241 & -189 & 40 \\ 8339 & 1270 & -269 \\ -1241 & -189 & 39 \end{pmatrix}. \tag{12}$$

These matrices satisfy $\det A = \det(A - I) = 1$. The sphere coordinate matrix in (7) is obtained by negating the columns of (11) and also has determinant 1.

Remark 6.1 (Long topology route)*.* Surjectivity of $A - I$ kills the natural coinvariant presentation of the fundamental group after surgery; bijectivity of $A - I$ and $(A^{-1})^T - I$ kills the intermediate homology in the Wang and Mayer–Vietoris sequences. Poincaré duality then gives the homology profile of S^4. Hurewicz and Whitehead promote simple connectivity plus that homology profile to a homotopy equivalence with S^4. This paper uses Iwaki Proposition 2.1 on the identified surgery manifold instead.

Corollary 6.2 (Homotopy-sphere conclusion for the identified picture)**.** *For the Johnson CS handle picture X_J identified with Σ_A^0 by P3/E13, Σ_A^0 is a homotopy* 4*-sphere.*

Proof. By (2), $A \in \mathrm{SL}(3, \mathbb{Z})$ and $\det(A - I) = 1$. The identification $X_J \cong \Sigma_A^0$ is the constructed CS handle picture; Iwaki Proposition 2.1 [12] then applies. □

6.3. **Coefficient traces and the selected Hattori class.**

Definition 6.3 (Coefficient trace)**.** For a small $\mathbb{Q}$-linear category $\mathcal{C}$ and coefficient bimodule M, the coefficient zeroth Hochschild homology is

$$\mathrm{HH}_0(\mathcal{C}; M) = \left(\bigoplus_T M(T, T) \right) \Big/ \mathrm{Span}_{\mathbb{Q}}\{f \cdot m - m \cdot f\},$$

where $f : S \to T$ and $m \in M(T, S)$ range over every cyclically composable pair.

Lemma 6.4 (Descent through the coefficient trace)**.** *Let $r_T : M(T,T) \to V$ satisfy $r_S(f \cdot m) = r_T(m \cdot f)$ for all cyclically composable f and m. Then there is a unique linear map $\bar{r} : \mathrm{HH}_0(\mathcal{C}; M) \to V$ whose restriction to $M(T,T)$ is r_T.*

Proof. The sum of the r_T kills every relation generator $f \cdot m - m \cdot f$ and therefore factors uniquely through the quotient. □

For the Johnson replacement collar, Lemma 10.1 supplies the actual coefficient equivalence (30). The selected class and divided detector of Section 7 are the geometric instances of the following pattern.

6.4. **The endpoint module and the Burau action.**

Definition 6.5 (Endpoint module)**.** Let $R_\varepsilon = \mathbb{Z}[\varepsilon]/(\varepsilon^7)$, $t = 1 + \varepsilon$, and $E_\varepsilon = R_\varepsilon^{88} = \bigoplus_{i=0}^{87} R_\varepsilon e_i$. One may identify this free module with the weight-86 subspace of the 88-fold tensor power of the two-dimensional fundamental module.

Definition 6.6 (Unreduced Burau representation)**.** For the Artin generator σ_i of B_{88}, with $1 \le i \le 87$, define $\rho_t(\sigma_i)$ to be the identity off the span of e_{i-1}, e_i and to have the block

$$\rho_t(\sigma_i)\big|_{\langle e_{i-1}, e_i \rangle} = \begin{pmatrix} 1-t & t \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -\varepsilon & 1+\varepsilon \\ 1 & 0 \end{pmatrix}.$$

Its inverse has block

$$\rho_t(\sigma_i^{-1})\big|_{\langle e_{i-1}, e_i \rangle} = \begin{pmatrix} 0 & 1 \\ t^{-1} & 1-t^{-1} \end{pmatrix},$$

where $t^{-1} = 1 - \varepsilon + \varepsilon^2 - \varepsilon^3 + \varepsilon^4 - \varepsilon^5 + \varepsilon^6$ in R_ε. For a word $w = \sigma_{i_1}^{s_1} \cdots \sigma_{i_m}^{s_m}$ we use the left action $\rho_t(w)v = \rho_t(\sigma_{i_1}^{s_1}) \cdots \rho_t(\sigma_{i_m}^{s_m})v$.

6.5. **From primitive crossings to the cabled word.** The public input stores 252 primitive crossing rows. For $i < j$ define the signed pure-row words

$$L_{ij}^s = \sigma_{j-1} \cdots \sigma_{i+1} \sigma_i^s \sigma_i^s \sigma_{i+1}^{-1} \cdots \sigma_{j-1}^{-1}, \tag{13}$$

$$R_{ij}^s = \sigma_i \cdots \sigma_{j-2} \sigma_{j-1}^s \sigma_{j-1}^s \sigma_{j-2}^{-1} \cdots \sigma_i^{-1}. \tag{14}$$

The 44-strand word B_{44} is obtained by taking the R rows on segment 6 in increasing horizontal coordinate, followed by the L rows on segment 2 in decreasing horizontal coordinate. The cabling homomorphism is

$$c(\sigma_i) = \sigma_{2i}\sigma_{2i+1}\sigma_{2i-1}\sigma_{2i}, \qquad c(\sigma_i^{-1}) = c(\sigma_i)^{-1}, \tag{15}$$

and the 88-strand word is $B_{88} = c(B_{44})$.

Proposition 6.7 (Word identities)**.** *The exact reconstruction gives $|B_{44}| = 11340$ and $|B_{88}| = 45360$. The Johnson reconstruction verifier recovers the same 11340-letter word letter-for-letter.*

6.6. **The detector and the exact cubic.** Let $R_h = \mathbb{Q}[h]/(h^7)$ and substitute $q = 1 + h$, $t = q^{-2}$, $\varepsilon = t - 1$. Under the endpoint normalization of the public input, the composite row is

$$\ell = e_{87}^* - e_2^*, \tag{16}$$

and the selected shadow vector is

$$u = e_2 - e_{87}. \tag{17}$$

Both formulas use the collar-coordinate endpoint table fixed in the public input.

Definition 6.8 (The divided cubic detector)**.** For a quantum trace class x, define

$$\mathcal{D}_h(x) = \widehat{C}(h)\big(\rho_h(W) - I\big)P_{86}\,\mathrm{Sh}_h(x),$$

where Sh_h is the quantum trace shadow, P_{86} projects onto the through-86 top cell, and $\widehat{C}(h)$ is the normalized endpoint cap row. When the image lies in h^3R_h, the divided cubic detector is

$$D_3(\bar{x}) = \left.\frac{\mathcal{D}_h(x)}{h^3}\right|_{h=0}. \tag{18}$$

Proposition 6.9 (Selected value)**.** *In E_ε, the first nonzero coefficient of $(\rho_t(B_{88}) - I)u$ occurs in degree* 3*. Contracting with* (16) *gives*

$$\ell(\rho_t(B_{88}) - I)u = -328\varepsilon^3 + 14596\varepsilon^4 - 410246\varepsilon^5 + 9595271\varepsilon^6. \tag{19}$$

Hence, with $[h]\varepsilon = -2$,

$$D_3([v_T]) = 2624 \neq 0. \tag{20}$$

The complete degree-three vector is printed in Appendix C.

Proof. Traverse B_{88} from right to left, reduce modulo ε^7, and contract with (16). The resulting series is (19), so $D_3([v_T]) = (-2)^3(-328) = 2624$. □

Lemma 6.10 (Parameter expansion)**.** *In $\mathbb{Z}[h]/(h^7)$, $\varepsilon = (1+h)^{-2} - 1 = -2h + 3h^2 - 4h^3 + 5h^4 - 6h^5 + 7h^6$.*

Lemma 6.11 (Commutator order)**.** *If $P = I + O(h^a)$ and $Q = I + O(h^b)$ are invertible matrices over an h-adically filtered ring, then $PQP^{-1}Q^{-1} = I + O(h^{a+b})$. If every generator of a pure-braid subgroup acts as $I + O(h)$, then its rth lower-central subgroup acts as $I + O(h^r)$.*

Remark 6.12 (Relative class)*.* The relative class $\xi = \eta_R[T_0] - s_{\mathrm{inv}}\eta_R[T_1]$ satisfies $D_3(\xi) = 0$ because the detector already contains one factor $\rho_h(W) - I$. The plain shadow coefficient $[h^3]\ell\,\mathrm{Sh}_h(\xi)$ equals 2624, but that plain shadow is not $D_3(\xi)$.

6.7. **The grading calculation.**

Proposition 6.13 (Quantum degree)**.** *For the Johnson replacement collar, the selected raw class has quantum degree* 494*.*

Proof. There are four contributions:

(21) $d_1 = -44$ (remove the 44 Hom normalizations for $N = 2$),

(22) $d_2 = +227$ (227 distinct Frobenius labels),

(23) $d_3 = +315$ (44 y-pairs and $271 = 44 + 227$ z-pairs),

(24) $d_4 = -4$ (cabled shift for $N = 2$, $\alpha = 0$, $|r| = 2$).

Adding gives $-44 + 227 + 315 - 4 = 494$. Appendix D collects the same table without abbreviation. ☐

6.8. **Quotient algebra and the abstract descent chain.** Let W_0 be the one-handle coefficient trace, W_1 its quotient by all two-handle relations, and W_2 the quotient of W_1 by the three-handle relations. Denote the selected class in W_0 by x_0.

Lemma 6.14 (A linear detector survives a quotient)**.** *Let V be a vector space, $R \subseteq V$ a subspace, and $\lambda : V \to \mathbb{Q}$ a linear map with $R \subseteq \ker\lambda$. Then there is a unique $\bar{\lambda} : V/R \to \mathbb{Q}$ such that $\bar{\lambda} \circ \pi = \lambda$. If $\lambda(x) \neq 0$, then $\pi(x) \neq 0$.*

Proposition 6.15 (Abstract nonzero class after quotients)**.** *Suppose Hypotheses 3.3 and 3.4 hold. Then there is a linear functional $\ell_2 : W_2 \to \mathbb{Q}$ such that $\ell_2(q_{12}q_{01}x_0) = 2624$. In particular, $q_{12}q_{01}x_0 \neq 0$.*

Proof. Lemma 6.4 descends the quantum shadow detector through the coefficient trace. Theorem 10.3 and Lemma 6.14 descend it through q_{01}. Theorem 11.4 and the same quotient lemma descend it through q_{12}. At every stage the pullback agrees with the preceding row, so Proposition 6.9 gives the value 2624. ☐

Proposition 6.16 (Closed candidate class)**.** *Suppose Hypotheses 3.4 and 3.5 hold and an inhabitant of Lean `ExternalGeometry` transports the class through the four-handle isomorphism. Then the class of Proposition 6.15 maps to a nonzero element of $\mathcal{S}_0^2(X_J)_{0,494}$. No such inhabitant is constructed.*

Proof. MWW Theorems 3.7 and 3.10 identify the three-handle quotient, and Proposition 3.4 identifies the four-handle stage once cited. Hypothesis 3.5 separates the computed empty-link ranks from those citations. The Lean implication in Theorem 3.1 is conditional on `ExternalGeometry`. ☐

Remark 6.17 (Presentation and serialization dependence)*.* The raw coefficient $[h^3]\ell(\rho_h(W)-I)u$ depends on the based, labelled, oriented movie presentation. Reordering the same crossing rows by the emitter's increasing-coordinate order produces a different series. With the corrected collar endpoints its cubic coefficient is 0 and its first nonzero coefficient is $663872h^4$. That order reverses 168 non-disjoint events and is not a legal serialization of the Johnson collar. The former value -58976 used the same mixed endpoint coordinates as the superseded public value. Presentation independence must be proved through simultaneous transport of every part of the naturality square; see Appendix E.

7. A fully worked finite-dimensional example

This section illustrates the algebraic mechanism on a small example. It is not a four-manifold calculation and supplies no evidence for any candidate-specific assumption.

Let $R_0 = \mathbb{Q}[\varepsilon]/(\varepsilon^4)$, $t = 1+\varepsilon$, and $E_0 = R_0^3$ with basis e_0, e_1, e_2. The two positive Burau generators are

$$B_1 = \begin{pmatrix} -\varepsilon & 1+\varepsilon & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad B_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\varepsilon & 1+\varepsilon \\ 0 & 1 & 0 \end{pmatrix}. \tag{25}$$

Take the pure commutator word $w_0 = \sigma_1^2\sigma_2^2\sigma_1^{-2}\sigma_2^{-2}$. Multiplying the eight matrices from left to right and reducing modulo ε^4 gives

$$\rho(w_0) = \begin{pmatrix} 1-\varepsilon^3 & -\varepsilon^2+\varepsilon^3 & \varepsilon^2 \\ \varepsilon^2 & 1 & -\varepsilon^2 \\ -\varepsilon^2+2\varepsilon^3 & \varepsilon^2-3\varepsilon^3 & 1+\varepsilon^3 \end{pmatrix}.$$

Let $u_0 = e_0 - e_1$ and $\ell_0 = e_0^*$. Define $d_2(v) = [\varepsilon^2]\,\ell_0(\rho(w_0) - I)v$. Then $d_2 = \begin{pmatrix} 0 & -1 & 1 \end{pmatrix}$ and $d_2(u_0) = 1$. Model two relation maps by the columns $r_0 = (1,0,0)^T$ and $r_1 = (0,1,1)^T$. Then $d_2 R = 0$, so d_2 descends to $E_0/\operatorname{im} R$ and the selected class is nonzero in the quotient. What this example lacks is precisely the geometric content: there is no claim that the toy relation columns arise from handle maps or that the final quotient is a smooth invariant.

8. Lasagna modules and the comparison framework

We recall only the features needed for the comparison. For a smooth compact oriented four-manifold W and a framed link $L \subset \partial W$, the skein lasagna module $\mathcal{S}_0^N(W; L)$ is generated by decorated surfaces in W, modulo local relations coming from KhR_N cobordism maps [18, 17].

For a handle decomposition

$$W_1 \subset W_2 \subset W_3 \subset W_4,$$

where W_1 is a one-handlebody, W_2 adds two-handles along a framed link, W_3 adds three-handles along embedded spheres, and W_4 adds four-handles, the published results have the following roles.

- MWW Theorem 3.2 identifies the two-handle stage with a cabled skein lasagna quotient, whose relations include beta and psi operations.

- MWW Theorems 3.7 and 3.10 describe each three-handle as a coequalizer of the two hemisphere maps, and give the complete handle formula.
- MWW Proposition 3.4 says that a four-handle attachment induces an isomorphism for the empty boundary link.
- MWW Corollary 3.5 gives $\mathcal{S}_0^N(S^4) \cong \mathbb{Z}$, concentrated in bidegree $(0, 0)$.

At $N = 2$, a nonzero class in bidegree $(0, 494)$ would therefore obstruct a diffeomorphism to S^4. The intended proof constructs a functional at the one-handle stage, shows that it annihilates the two-handle and three-handle relations, and transports the resulting nonzero class through the four-handle isomorphism.

The formal algebra correctly captures the last sentence. The next sections construct and verify the geometric identifications required for P0, C and S.

9. Johnson product-ribbon presentation and P0

We record the AR-side construction, summarize rejected compact lifts, and give the Johnson-generator replacement that proves Hypothesis 3.2.

9.1. **The actual AR link.** Use the coordinate-spine genus-three Heegaard splitting of T^3 from Aitchison–Rubinstein [1, pp. 5–12]; the complete volume is available in a public Berkeley scan. Let C_i be the three coordinate spine circles with common base vertex Q. In the notation of their p. 6, choose the nested section balls $R' \subset R$ and put

$$K_i = C_i - (R - \operatorname{int} R').$$

Thus K_i is the cut coordinate arc used in the handle construction, not the whole based circle. Let A_i be its narrow disk neighborhood. Write t for the mapping-torus one-handle. If ψ_A is the handlebody-preserving representative supplied by their Lemma 3.1, the core of the ith mapping-torus two-handle is the embedded circle

$$K_i^- \cup \lambda_i \cup \psi_A(K_i)^+ \cup \mu_i, \tag{26}$$

where λ_i, μ_i are the two arcs through t. The framing annulus is constructed simultaneously as

$$A_i^- \cup (\lambda_i \times I) \cup \psi_A(A_i)^+ \cup (\mu_i \times I). \tag{27}$$

The disks, cones and product levels in the AR construction are pairwise disjoint, so (26)–(27) are the data that must be reproduced, not merely their words. The current rational artifact cuts at radius $1/392208$ inside the pointwise-fixed ball, uses the actual lane-spine arc for every $\psi_A(K_i)$, and closes its distinct endpoints by the two mapping-handle arcs. The three embedded centre lines pass their live rebuild and endpoint-mutation checks. The constant rational product direction $(1, 1, 1)$ is transverse to every edge of these cores and to the square of every one of the 93 Johnson slides: its dot product with each recorded square normal is nonzero. If $D = 392208$

is the largest reduced denominator among all core coordinates, thickening by width $1/(100D^{64})$, strictly inside both the disjoint lane tubes and the fixed cut ball, therefore gives explicit two-triangle rectangles along the entire core. The same product direction across every slide prism records relative twist zero. This closes the transported top ribbons $\psi_A(A_i)$ and the actual framed-link gate.

The remaining three fiber two-handles are the standard dual two-cells of the coordinate-spine splitting. After choosing orientations, denote their boundary curves by

$$r_{xy} = [x, y], \qquad r_{yz} = [y, z], \qquad r_{zx} = [z, x].$$

Their embeddings are defined by the dual cells and their framings are the fiber product framings. The untwisted section-surgery handle h_{CS} goes geometrically once over t. Thus the starting handle counts are $(1, 4, 7, 3, 1)$ in indices $0, \dots, 4$: AR p. 8 removes one 3- and the 4-handle from the mapping torus and glues $S^2 \times D^2 = H^2 \cup H^4$ in their place.

For an independently checkable standard-form bridge, set

$$C_{AR} = \begin{pmatrix} 0 & 0 & 1 \\ 713 & 83 & 0 \\ -902 & -105 & -10 \end{pmatrix}, \qquad R = \begin{pmatrix} 71 & -466 & 1 \\ -610 & 4004 & -7 \\ 1 & -7 & -2 \end{pmatrix}.$$

Direct multiplication gives

$$\det R = 1, \qquad C_{AR}R = RA,$$

and both matrices have determinant one and determinant-one difference from the identity. Hence the fiber diffeomorphism induced by R transports the complete AR construction to the displayed matrix A. Since $R \in GL^+(3, \mathbb{R})$ and this group is connected, its derivative preserves the homotopy class of the canonical normal framing of the section circle; in particular, the untwisted choice $\varepsilon = 0$ is preserved.

9.2. **Return to the linear torus map.** Let ϕ_A be the linear torus map, straightened to the identity on the section ball, and choose an isotopy ρ_u, $u \in [-1, 1]$, from ϕ_A to ψ_A, stationary near its endpoints and relative to that ball. Put $g_u = \rho_u \phi_A^{-1}$. With mapping-torus convention $(x, -1) \sim (\phi(x), 1)$, the formula

$$F([x, u]_{\phi_A}) = [g_u(x), u]_{\psi_A} \tag{28}$$

defines a diffeomorphism. Indeed, $g_{-1} = \mathrm{id}$, $g_1 = \psi_A \phi_A^{-1}$, and therefore $\psi_A g_{-1} = g_1 \phi_A$, which is precisely well-definedness at the seam. Endpoint stationarity makes (28) smooth, and the family g_u^{-1} gives its inverse. It fixes the section ball. Pulling the entire AR handle decomposition back by F replaces the top cores by the linear images $\phi_A(C_i)$ and transports every component and framing annulus simultaneously.

9.3. **Words and normal fields.** Lift C_i and $\phi_A(C_i)$ to the universal cover. The three top lifts begin at the same point $A(Q)$. Translate the whole top fiber by $-A(Q)$ and extend this translation over its collar. It is isotopic to the identity and transports the entire link and its normal fields. Since each attaching core is unbased, changing the first face crossing only cyclically permutes its word. We may therefore take the top centerline to be the straight segment from 0 to Ae_i; it crosses the jth integer faces at the rational times $k/(Ae_i)_j$. A fixed arbitrarily small product perturbation orders simultaneous events. Reading the two base arcs and the oppositely oriented bottom core gives exactly

$$m_i = t\phi_A(x_i)t^{-1}x_i^{-1}. \tag{29}$$

This is the full event rule used by the public generator.

The framing is equally explicit. AR push the coordinate axes in the universal cover in direction $(1, 1, 1)$; the resulting strips have parallel boundary components, and linearity of A preserves this property on their images [1, pp. 16–17]. Together with the λ_i/μ_i rectangles, these strips are the normal fields in (27). Thus no blackboard integer is substituted for a framing.

9.4. **The two cancellations.** Aitchison–Rubinstein identify (t, h_{CS}) as a complementary geometric one/two-handle pair [1, p. 7]. Slide every other two-handle passage over h_{CS} along its product rectangle and cancel; this removes the t, t^{-1} passages with zero relative twist. In the actual octahedral belt chart the seven passage points are the h_{CS} point and the six signed endpoints of the three cut spine arcs. Six sequential rational bands on the belt sphere use distinct parallels of h_{CS} and remove precisely the latter six passages before cancellation.

Now $Ae_1 = e_3$, so the actual curve m_1 becomes zx^{-1} and has one geometric passage over the x one-handle. Thus (x, m_1) is another complementary pair. Sliding every other x passage over m_1 along parallel product bands replaces x by z and cancels the pair. All labels and normal fields move with the bands. The cubical x-belt chart contains one cancelling m_1 point, 269 actual lanes from m_2, 1240 from m_3, and two points from each of r_{xy} and r_{zx}. Thus the live movie has 1513 sequential zero-twist bands; every band is attached to its lane or dual-cell vertex identifier and replaces it by the equally oriented parallel of the actual top z-lane of m_1. The resulting counts are $(1, 2, 5, 3, 1)$, and the exact word projection is

$$|m_2| = 311, \quad [m_2]_{(y,z)} = (40, 269),$$
$$|m_3| = 1460, \quad [m_3]_{(y,z)} = (189, 1271).$$

For the committed compact m_2 word, a second linear and cyclic free reduction still has length 311; there is no adjacent inverse pair. The explicit 19-step Nielsen representative constructed in Appendix B instead reduces to length 309. It contains 40 y/Y passages, rather than 42; with the two r_{xy} passages its detector has 42 channels rather than 44. Thus the 309/311 discrepancy is a genuine representative discrepancy, not merely a printing convention. The transported r_{zx} has empty free word. That is a word-level fact only: it does not determine an embedded split unknot.

9.5. **The actual detector ball.** Cut the remaining genus-two handlebody along its y, z disk system. The selected m_2/r_{xy} state has 42 y passages from m_2 and two from r_{xy}. These are now cut directly from the post-cancellation event lists: each of the 44 height-monotone arcs records its original lane or dual-cell identifier, orientation, paired z event and transported product normal; the complementary list has exactly 227 unpaired z circles. Their distinct rational belt points lie in an explicit PL three-ball B containing a punctured-disk collar. A 44-step point motion, one point at a time and missing all stationary points, identifies those actual points with normalized lanes; its reverse is the endpoint-return chart. The six affine sweeps specify a pure 44-strand braid in this collar. The mapping-class interpretation of the braid group realizes it by an isotopy of the punctured disk, and isotopy extension gives an ambient isotopy of B. Choose the extension to be the identity to first order near every returned puncture at its terminal time. Hence each product normal returns, not merely the total writhe. The motion acts on every oriented normal copy simultaneously. Reconstruction of the 252 crossing factors gives $|B_{44}| = 11340$ with $\mathrm{perm}(B_{44}) = 1$ and $\mathrm{wr}(B_{44}) = 0$. The public word is read only after this PL motion has been built and its 11340 crossings re-extracted; it is a terminal comparison, not a construction input.

Remark 9.1 (Rejected compact lifts). Earlier symbolic lifts, including a 19-step Nielsen representative and bounded IA corrections, fail channel count, free-basis, or embedded-collar tests. Appendix A records the falsified routes.

9.6. **Johnson replacement.** Johnson's splitting-preserving generators for the genus-three splitting of T^3 [13] give a more appropriate search space. His generator α_{ij} pushes the diagonal of the x_i–x_j square to a neighborhood of the spine; pushing to the opposite side gives a second lift of the same transvection. Factoring A into 93 unit transvections and choosing a 93-bit side pattern yields

$$|m_2| = 311, \qquad \#(y/Y) = 42, \qquad p_y = 42 + 2 = 44,$$

with the three spine images forming a free basis of F_3 and the reduced m_2 word agreeing with the compact word. The class-two r_{yz} coefficient relative to the compact word is zero.

For each square move we record the diagonal, a two-edge path, and a square normal. Truncating both paths at one-quarter and three-quarters makes the movie relative to the spine vertex. If M ranges over the 93 intermediate unimodular bases, the uniform radius

$$r = \frac{1}{8 \max_M \|M^{-1}\|_\infty} = \frac{1}{196104}$$

is fixed throughout. A fixed-boundary Freudenthal cutoff on a cube of half-side $6r$ cancels each affine transvection on the image of the protected r-ball. Its 162 rational tetrahedra have positive Jacobians and cellwise inverses, so the composed representative is pointwise the identity on that ball. This local cutoff does not repair the Heegaard block mismatch on the coordinate arms; the required Johnson restore map and setwise preservation of the splitting were still open at this intermediate cutoff stage; the later Johnson restore assembly closes them. The live gate tests both owners at all 64 unit-cube centres and all 384 Freudenthal tetrahedron barycentres. For the present composite it finds 32 and 128 mismatches, respectively, so no Heegaard-preserving claim is inferred from the local cutoff. For the arm correction, exact rational half-space clipping decomposes every unwrapped affine image tetrahedron against the target unit cubes. Up to permutation of the coordinate axes, a positive unit transvection has 192 nondegenerate mismatch polytopes and a negative one has 96. In each case the two owner directions occur equally often. None meets the eight-cube star of the origin. These polytopes specify the required support of the Johnson restore, but no fixed-boundary homeomorphism of that support is yet claimed. Exact common-face gluing on the torus splits either signed mismatch into six connected components, three in each owner direction. Every component has a connected closed boundary with edge multiplicity two and Euler characteristic 2. Thus the boundary-sphere gate passes; the interior-link gate is resolved by a common-face cone triangulation, and an explicit free-face sequence collapses every component to a point. Each lifted component lies in a box of coordinate span at most $2 < 4$, so these are actual embedded PL 3-balls rather than balls created by quotient identifications. Pairing the three excess balls with the three deficiency balls by the selected Johnson-side bands is unique at the polyhedral level: each deficiency ball, including its full rational subdivision, is exactly the corresponding excess ball translated by half a period along the prefix axis. A rigid half-turn does not respect the nonsymmetric subdivisions. At this intermediate point, constructing disjoint Johnson-side exchange collars and their fixed-boundary PL transports was the next unresolved step; the later assembly supplies it. Each of the 72 signed, oriented component occurrences also has a strict rational star centre. Coning its boundary triangulation to that centre and using the homothetic scales $1/4, 1, 3/2$ gives a fixed-boundary radial shrink in the outer $3/2$-ball. The 8448 resulting affine tetrahedra have Jacobians between $1/64$ and $5/2$, and swapping source and image tetrahedra gives the explicit inverse. Thus shrinking and re-expanding the mismatch balls is certified in each individual ball chart. The six closed balls meet along a seven-edge intersection graph, and half-period translation preserves that graph. Hence the individual shrink charts cannot yet be composed into one ambient map; a compatible movie on the common edges and vertices is required. The exact boundary patches determine that movie's Morse skeleton: four small balls carry disk-to-disk moves, one large ball carries an annulus-to-two-disks saddle, and the paired large ball carries the inverse two-disks-to-annulus saddle. This $4+1+1$ pattern is the same in all twelve signed axis templates and is checked from the source and target owner changes on every boundary facet. At the level of scale-$1/4$ interior cores, support scale $3/8$ leaves enough periodic clearance: the large pair uses opposite detours in the third axis and the two small pairs use outward detours in the source axis. Exact swept-box tests for all three path phases and all twelve signed axis templates find no support-interior collision and no protected-ball hit. The routes are therefore fixed; attaching positive-Jacobian compact translation cells along them and joining them to a common-intersection shrink movie was the next intermediate gate, closed by the later ambient-cell assembly. The complete affine/cubical overlay makes the required side choice finite. For positive unit transvections it has 3584 tetrahedra grouped into 192 mandatory-transport cells, 192 mandatory-fixed cells and 384 flexible cells; the negative template has half as many polyhedral cells and respective counts $96, 96, 192$. A moving cutoff that changes only the prefix coordinate can transport every mismatch vertex exactly. Exact positivity forces 17 steps in the positive template and 9 in the negative template, with minimum step Jacobian $1/17$ and $1/9$. Nevertheless both leave 16 cube-centre and 96 tetrahedron-barycentre owner errors. This fiber-preserving route is therefore rejected: the restore must use transverse Johnson-square motion in the flexible cells. The common overlay also admits a fully checked combinatorial sweep. Every single-tetrahedron move is accepted only when its attachment is a nonempty proper disk and the updated boundary remains a connected genus-three two-manifold with both sides face-connected. The negative prefix-first and target-first movies use respectively 302 and 304 such moves; both positive movies use 794. Each movie then has exactly two residual collapsible balls: an add-ball attached along two disks and a remove-ball attached along an annulus. Treating these together as one paired saddle leaves no residual tetrahedron and reproduces the target handlebody simplex by simplex, with boundary Euler characteristic -4. Turning each recorded disk move and the paired saddle into explicit ambient PL cells with positive Jacobian was the next intermediate step and is completed below. The local ambient cell problem for every disk move has a uniform solution. After adding the midpoint of the exchanged edge, the refined tetrahedron boundary is an octahedron; both the $1 \leftrightarrow 3$ move and the $2 \leftrightarrow 2$ move are conjugate to its half-turn. Use four rational angular sectors and octahedral radii $1, 1001/1000, 501/500$. The two Johnson sides use sector shifts $(2, 1, 0)$ and $(2, 3, 0)$, respectively. A compatible staircase triangulation gives $8 + 24 + 24 = 56$ affine tetrahedra per side, all with Jacobian exactly 1, a fixed outer boundary, and explicit cellwise inverse. Instantiating the template at all 2194 recorded disk moves gives 122864 affine cells with Jacobians in $[1/3, 3]$. Every instantiated source and image is face-to-face, its eight outer triangles agree pointwise, and the minimum periodic bounding-box clearance from the protected ball is $7989/8000$. Thus all single-move ambient collars are complete; constructing the paired-saddle collar was the remaining intermediate task at this stage. For the final collar, the shortest dual-face path gives 85- and 83-tetrahedron balls for the two negative sides. In the positive cases the shortest path has an extra dual cycle; requiring an induced path that meets each residual ball through exactly one face gives a 20-step path and a 122-tetrahedron ball for either side. All four supports have manifold sphere boundary, collapse to a point, and miss every protected-ball centre by at least distance 1. The current and target surfaces cut each support in a single disk. In a negative support the two boundary polygons share one edge and cut the support sphere into three disks. In a positive support the two triangular boundaries are disjoint and their complement is an annulus and two disks. Thus the paired-saddle ball and its disk-transition pattern are certified. The outer boundary must carry the corresponding disk half-turn; placing that boundary map and its fixed-boundary polar collar was the next intermediate gate. The path itself has a canonical manifold thickening: alternate tetrahedron and shared-face barycentres, include the two cap-face barycentres, subdivide twice, and take the closed star of the subdivided core arc. For a negative move this star has 648 tetrahedra and boundary data $(V, E, F) = (266, 792, 528)$; for a positive move it has 4320 tetrahedra and $(V, E, F) = (1694, 5076, 3384)$. In both cases every boundary edge has multiplicity two, the boundary has Euler characteristic 2, and an explicit incremental free-face sequence collapses the star to a point. Thus the cap-to-cap $D^2 \times I$ regular neighbourhood is certified; its fiber-reversing PL map is the remaining paired-saddle cell problem. The standard fiber-reversing cell map is already explicit. Inside the unit octahedron take the diamond fibers at $x = -1/2$ and $x = 1/2$. The inner half-turn $(x, y, z) \mapsto (-x, -y, z)$ carries the first four-triangle disk onto the second, while the two angular staircase shells make the map pointwise identity on the outer octahedron. For either Johnson side this uses the same 56 determinant-one cells and their recorded inverses. What remains is an explicit orientation-preserving PL chart from each actual second-derived path star to this standard fiber model. Cutting each of the four paired-saddle balls along its source disk and again along its target disk gives sixteen side balls. Each side has an explicit relative elementary-collapse sequence onto the whole cutting disk, with all vertices, edges and triangles of that disk protected. Across the sixteen cases there are 2604 collapse pairs; after replay, the remaining simplex set in every dimension agrees exactly with the downward closure of the protected disk. Thus both sides of every actual disk are certified regular neighbourhoods of that disk. Expanding these collapse pairs into positive-Jacobian derived-star ambient cells is the remaining chart step. There are only three local derived-star types. In the twice barycentrically subdivided standard tetrahedron, the before/after stars for collapse dimensions $1, 2, 3$ contain respectively [(96,36),qquad(216,156),qquad(576,396)] tetrahedra, with differences $60, 60, 180$. Every before and after star has a manifold sphere boundary and collapses to a point. Removing their common boundary leaves a disk on either side; the common boundary polygons have $6, 30, 12$

edges, respectively. All rational second-derived tetrahedra are nondegenerate. These are the three finite local inputs from which the 2604 actual collapse-pair charts must be instantiated; their ambient cell maps are obtained by coning each difference sphere to the recorded strict rational star centre. Sweeping first across the before-disk cone and then across the after-disk cone gives respectively $72, 72, 168$ disk moves in dimensions $1, 2, 3$. For both Johnson sides this totals 624 moves and 34944 actual octahedral cells. Every intermediate interface is recomputed to be a disk; all Jacobians lie in $[1/3, 3]$, and every cell has the recorded inverse and fixed outer boundary. The three standard derived-star ambient maps therefore pass. Their placement at the 2604 actual collapse pairs is also finite. For a dimension-d pair (τ, σ), map standard vertex 0 to $\tau \setminus \sigma$, vertices $1, \ldots, d$ to the free face, and the remaining vertices to a carrier tetrahedron containing τ. Exhausting the finite vertex orders and retaining only positive determinants gives an orientation-preserving affine chart for every actual pair. The charts and their affine inverses are checked on all four vertices. Conjugating the standard maps yields 2604 placements representing 14929152 positive cells with Jacobians in $[1/3, 3]$; every expanded carrier box misses the protected ball. Swapping two chart-image vertices makes the first determinant negative, so the orientation mutation fails. Thus all side-chart cells pass. The fiber transport between the source and target disks was the remaining task here; the later restore assembly supplies it. The two endpoint fibers of each second-derived path star are detected directly from their original-face carriers. Each is a 12-triangle fan disk. The negative path star collapses relative to either cap in 1812 steps, with dimension histogram $(648, 900, 264)$ in dimensions $(3, 2, 1)$; the positive path star collapses relative to either cap in 12080 steps, with histogram $(4320, 6000, 1760)$. Replay leaves exactly the chosen cap in every dimension. Thus charts from the actual tube to either endpoint fiber are now finite and symmetric. Each cap has exactly one interior fan vertex and a twelve-edge boundary cycle. Mapping fan centre to fan centre and the boundary cycles in either cyclic orientation gives two explicit simplicial disk isomorphisms; both are checked triangle by triangle. The remaining orientation choice must be made by the three-dimensional tube-chart determinant, after which the cap-collapse charts can be composed. An actual one-sided product collar removes this last ambiguity. Move every cap vertex by $1/1000$ of the segment toward the adjacent path-tetrahedron centroid and triangulate each of the twelve triangular prisms by the common staircase rule. In the negative templates the preserving fan map gives 36 cells of determinant 1, while the reversing map gives determinant -1. In the positive templates the reversing fan map gives determinant $6/5$, while the preserving map gives $-6/5$. Thus the unique orientation-compatible cap map is selected live in all four movies; its product cells are face-to-face and have explicit inverses. Rebuilding the relative collapse sequence on every second-derived path star gives 55568 actual cap-collapse charts. Each is placed in a positive carrier and checked with its affine inverse. In the order

$$H_{\mathrm{cap0}}^{-1} \circ C_{\mathrm{cap}}^{-1} \circ H_{\mathrm{cap1}}$$

the four movies expand to $21579300, 21579300, 143861796, 143861796$ cells, respectively, for a total of 330882192. Their Jacobians lie in $[1/3, 3]$ and the inverse is the recorded reverse composition. Thus the internal fiber transport passes. Its boundary action has not yet been joined to the 14929152 paired-support side-chart cells to give a map fixed on the outer support boundary; the paired-saddle ambient map was still an intermediate gate. Here cap 1 is the current/remove endpoint and cap 0 is the target/add endpoint in every movie; the inverse collar direction is therefore essential. For positive moves these caps are exactly the source and target disks. For negative moves the source disk is cap 1 together with one adjacent triangle, so a two-step relative disk normalization is still required before the final paired-saddle assembly. That normalization is the relative two-dimensional collapse of the extra triangle onto cap 1: first a dimension-two triangle/free-edge pair, then a dimension-one edge/free-vertex pair. Each negative movie therefore uses two positive actual carrier charts and 8064 expanded cells; the two positive movies require no normalization. All four charts have explicit inverses, so the source-disk endpoint now agrees exactly with cap 1. Joining this normalization and the internal fiber transport to a map fixed on the outer paired-support boundary was the remaining local gate at this stage. On the support sphere, the source and target boundary curves differ by a unique finite triangle region: a five-triangle disk in each negative movie and a forty-triangle annulus in each positive movie. XOR-ing the triangle boundaries in the certified order keeps a single simple closed curve at every stage and ends at the target curve. Thickening each $1 \leftrightarrow 2$ edge move by an outward point at scale $1/1000$ and conjugating the standard dimension-two map gives 90 actual collar moves and 362880 positive cells. All carrier charts and inverses are explicit, and the expanded carrier boxes miss the protected ball. Hence the outer boundary extension passes.

Combining the single disk moves, source side charts, negative normalization, cap-1 to cap-0 transport, inverse target side charts, outer curve collar, and section cutoff gives four canonical unit restores with 346313864 expanded cells. Axis-permutation conjugation places them at all 93 Johnson factors; the resulting hierarchy represents 8113517822 cells, fixes the protected ball, maps both owners setwise, has explicit reverse-order inverse, and its transvection product is A. Changing the first side bit is detected. The hierarchical restore assembly therefore passes. The version-two generator serialization removes the legacy square-fan map and records $\mathrm{ArmRestore} \circ \mathrm{SectionRestore} \circ A_{ij}$ at all 93 factors. Its live verifier reports zero owner mismatch at all 64 cube centres and 384 tetrahedron barycentres, fixes the protected ball, and recomputes the product on H_1 as A. Thus the setwise Heegaard-preserving PL homeomorphism is complete. At this intermediate stage the hierarchy had not yet been evaluated on the actual spine curves; the later lane-spine binding and rebuilt AR link close that gate and reject the old AR-link SHA. The target spine itself is now embedded independently of the old point evaluator. The three Johnson generator images have word lengths $(1, 310, 1461)$ and 1772 total handle traversals. Every occurrence is placed in a distinct rational lane of its coordinate one-handle. The 1775 successive port pairs are joined in the origin zero-handle by distinct height levels: vertical columns have distinct xy projections, and each horizontal two-segment path is checked against every other column projection. The three closed PL components meet only at the fixed origin and agree with the original axis spokes on the protected ball. Retraction reads the three generator-image words and abelianizes to A. Duplicating a lane, duplicating a connector height, reversing a letter, or changing the first Johnson side bit is detected. Thus the actual lane spine is embedded; binding it by an explicit sequence of the 93 ambient Johnson handle slides was the next gate, discharged by the lane-spine binding below.

In the standard product chart $D^2 \times [-1, 1]$ for the reduced y one-handle, take $B = D^2 \times [-1/2, 1/2]$. The Johnson m_2 word has 41 positive y passages and one negative passage; $r_{xy} = [z, y]$ contributes one positive and one negative passage. Distinct rational points in D^2 give pairwise-disjoint product arcs with constant transverse framing. The oriented dual rectangle for r_{xy} has six point-push legs; expanding the resulting 252 geometric L/R factors gives an Artin word of length 11340, which agrees with the independently regenerated public word.

Lemma 9.2 (Railroad linking)**.** *The railroad reduced planar diagram of the Johnson CS handle picture satisfies* $\operatorname{lk}(m_2, r_{yz}) = 0$.

Proof. The mixed m_2–r_{yz} signed crossing sum in the labelled reduced PD is 0, so the linking number vanishes. By the Kirby handle-slide formula [15], this is the framing condition required for the product cancellations above. □

9.7. **The three geometric identifications.**

Lemma 9.3 (Johnson–AR handlebody bridge)**.** *The affine map*

$$S : \mathbb{R}^3/\mathbb{Z}^3 \longrightarrow \mathbb{R}^3/(2\mathbb{Z})^3, \qquad S(u) = 2u - (1/2, 1/2, 1/2),$$

is an orientation-preserving diffeomorphism carrying Johnson's standard genus-three Heegaard pair to the coordinate-spine pair used by Aitchison–Rubinstein.

Proof. The period-four cubulation of T^3 has a $2 \times 2 \times 2$ coarse cellulation. The barycentric dual 3-blocks of the four vertices of each coordinate spine are genus-three handlebodies: each collapses onto its spine by an explicit free-face elementary-collapse sequence, the two blocks meet in a closed genus-three surface and fill T^3, and Regina's `recogniseHandlebody` returns genus 3. Tetrahedra are assigned as whole dual 3-blocks of cubical vertices, not by comparing tetrahedron barycentres to the spines. In the integer model

$T(v) = v - (1,1,1)$ of S, this Johnson dual-block pair is carried simplex-by-simplex onto the AR dual-block pair. □

Lemma 9.4 (The two framed cancellations)**.** *In the Johnson–AR replacement, the pairs* (t, h_{CS}) *and* (x, m_1) *satisfy the geometric Kirby* $1/2$*-cancellation criterion. The cancellations transport the whole labelled attaching link and preserve the section framing class* $\varepsilon = 0$*.*

Proof. For the selected Johnson lift, $\psi_A(x) = z$ and $m_1 = zx^{-1}$. Each pair admits a local product 1/2-handle cancellation of geometric intersection 1 that transports the labelled attaching link and preserves $\varepsilon = 0$. □

Lemma 9.5 (Compatibility with MWW cabling)**.** *The framing on the 44 Johnson lanes is the framing used by the MWW cable construction.*

Proof. The 44 Johnson six-sweep strands are realized as height-monotone polylines in a triangulated 3-ball with constant product normals, bound to the Johnson y-wickets in the certified handlebody pair. Reconstruction recovers the public 11340-letter word. □

9.8. **P0 conclusion.**

Theorem 9.6 (P0 status for the Johnson replacement)**.** *Hypothesis 3.2 holds for the explicit Johnson presentation.*

Proof. The Heegaard-preserving 93-factor map, seven-component framed AR link, six and 1513 Kirby bands, actual 44-channel cut, endpoint-return chart, and terminal word comparison supply the required bridge, cancellations and cabling data. The E13 chain identifies the resulting closed picture with Σ_A^0. □

10. The coefficient-trace comparison

This section records the C argument for the Johnson replacement collar.

10.1. **The product Hattori equivalence.** The compact words give the letter counts

$$p_y = 42 + 2 = 44, \qquad p_z = 269 + 2 = 271.$$

In the cyclic m_2 word, every one of the 42 y/Y events is followed immediately by a distinct z/Z event; the same holds for the two y/Y events of r_{xy}. The witness lists all 44 pairs. Those pairs are realized as 44 product ribbons of the P0 reconstruction strands along their certified normals, with 227 leftover z-circles off the P0 cube. The scope is the Johnson replacement collar, not an embedded cut in ∂W_2.

Let $\mathcal{C}$ be the pregraded rational tangle category with objects $T : P_{86} \to P_{88}$, and let F be the framed west-to-east tangle formed by those rectangles. MWW's one-handle theorem identifies the cut coefficient as

$$M_R(T, T') = \mathrm{KhR}_2(R \cup T' \cup \overline{T})$$

with its displayed shift and gluing actions [17, Theorem 4.7]. Pivotal tangle duality and Künneth over $\mathbb{Q}$ give graded isomorphisms on this replacement collar

(30)
$$H_{T,T'} : M_R(T,T') \xrightarrow{\cong} \mathrm{Hom}_{\mathcal{C}}(FT, FT')\{-44\} \otimes A^{\otimes 227}, \qquad A = \mathbb{Q}[X]/(X^2).$$

These maps form a coefficient-bimodule isomorphism on the replacement collar. Gluing $f : S \to T$ before the source link and then applying the product isotopy is isotopic rel boundary to first applying the isotopy and precomposing by Ff. The right action is the same argument with postcomposition. Thus both action squares are associativity of gluing for one simultaneous surface. At the chain level the two squares are

(17)
$$\begin{array}{ccccccc} C_R(T,T') & \xrightarrow{H_{T,T'}} & C(FT,FT') \otimes C(S^1)^{\otimes 227} & \qquad & C_R(T,T') & \xrightarrow{H_{T,T'}} & C(FT,FT') \otimes C(S^1)^{\otimes 227} \\ \downarrow L_f & & \downarrow L_{Ff} \otimes 1 & & \downarrow R_g & & \downarrow R_{Fg} \otimes 1 \\ C_R(S,T') & \xrightarrow{H_{S,T'}} & C(FS,FT') \otimes C(S^1)^{\otimes 227}, & & C_R(T,U) & \xrightarrow{H_{T,U}} & C(FT,FU) \otimes C(S^1)^{\otimes 227}. \end{array}$$

Here C_R and C are the strict Blanchet–Khovanov foam complexes. Both composites in each square are the same glued foam after exchanging two disjoint collars. BHPW strict functoriality removes the projective sign, and taking homology gives (30) and both MWW action squares for the Johnson replacement collar. Equation (30) is not a chain-level complex of an embedded cut in ∂W_2.

The ordinary quotient statement, including both cyclic-relation spans, is checked in the formal development.[2]

Lemma 10.1 (Actual coefficient bimodule)**.** *For the P0 replacement,* (30) *is a natural isomorphism of the actual MWW coefficient bimodule, with both gluing actions. Moreover the map* Sh_q *below is the quantum trace of that bimodule, not a dimension-matched substitute.*

Proof. The 44 product ribbons are realized from the P0 reconstruction strands with certified product-normal z-sides, giving the Johnson y-then-z pairing and 227 leftover z-circles. The action cubes of (30) and the representable HH_0 reduction of Section 6 complete the comparison. □

10.2. **Quantum trace and the endpoint map.** Set $R_q = \mathbb{Q}[q, q^{-1}]$. The pregraded cyclic relation is

$$L_f(m) = q^{|f|} R_f(m).$$

By Lemma 10.1, (30) and the counits are homogeneous coefficient morphisms on the Johnson replacement collar. BPW's canonical vertical-to-horizontal functor and the BHPW strict tangle/foam functor give

(31) $$\mathrm{Sh}_q : q\,\mathrm{Tr}(\mathcal{C}; M_R) \longrightarrow \mathrm{Hom}_{R_q}(E_{86}, E_{88}),$$

---

[2] See `Smooth4PC/RepresentableCoefficient.lean`.

after the flat-base qHH_0/Chern identification [4, 3]. Here

$$E_{86} = (V^{\otimes 86})_{\lambda=86}, \qquad E_{88} = (V^{\otimes 88})_{\lambda=86};$$

the first has rank one and the second rank 88. The tangle maps are $U_q(\mathfrak{sl}_2)$ intertwiners before restriction to these weight spaces.

Base change $q \mapsto 1+h$ factors through rational localization and the completion of a Noetherian local ring, hence is flat. Reduction modulo h of the trace cokernel simply replaces $q^{|f|}$ by one, giving the ordinary MWW coefficient HH_0; right exactness of tensor product suffices for this last assertion.

10.3. **Selected class and divided detector.** Let $U : P_{86} \to P_{88}$ be the selected one-cup tangle in the P0 endpoint order and put $T_1 = F^{-1}U$. Define

$$v_T = H^{-1}_{T_1,T_1}(\mathrm{Id}_U \otimes X^{\otimes 227}).$$

The circle counits give $\mathrm{Sh}_h([v_T]) = u_h$. Its absolute degree is

$$-44 + 227 + 315 - 4 = 494.$$

Before any public index is assigned, the 44 actual product rectangles give 88 physical endpoints: the positive side records the actual y source and the negative side its paired z source. Boundary orientation orders the two r_{xy} wickets forward and the 42 m_2 wickets in reverse, with negative then positive side in each rectangle. Only then does the public position table attach Burau indices. The resulting monomial map $P(q)$ and its inverse are recomputed from these physical identifiers; all pivotal coefficients are $+q^0$, and sign, order, and pivotal mutations fail.

The normalized checked R-matrix on adjacent one-defect basis vectors is

$$\begin{pmatrix} 1-q^{-2} & q^{-1} \\ q^{-1} & 0 \end{pmatrix}.$$

After the position basis change and $t = q^{-2}$, it is the public positive Burau block; the negative block is its inverse. The physical cable has mixed orientations. BPW's oriented-cabling formula identifies it with the all-positive model up to a writhe-dependent grading shift and pivotal basis maps. The complete public word has writhe zero, so the global shift vanishes. Any degree-zero permutation or pivotal chart P transports the operator, cup and cap simultaneously:

$$W \mapsto PWP^{-1}, \qquad u \mapsto Pu, \qquad \ell \mapsto \ell P^{-1}.$$

The matrix coefficient is unchanged; we choose the public endpoint labels after this common transport. Unresolved $\pm$ formulae for the constant terms are not used. In the transported collar coordinates of Section 6 the constant pairing is the definite identity recorded there, not a sign ansatz.

Over $\mathbb{Q}[[h]]$ define

$$D_h = \ell_h(\rho_h(W) - I)\,\mathrm{Sh}_h\,.$$

This is well defined on the quantum coefficient trace because (31) has already imposed quantum cyclicity. The exact Magnus calculation and Darné's Andreadakis theorem give $W \in \Gamma_3(P_{44})$ [6, Theorem 6.2]. This implies

$$\rho_h(W) - I \in h^3 \operatorname{End}(E_{88}).$$

An independent verification evaluates all 7,744 matrix entries of the truncated Burau representation directly; that computation is not treated as equivalent to Andreadakis membership, but as a separate check of the same cubic vanishing. Thus D_h takes values in $h^3\mathbb{Q}[[h]]$. Division by h^3 is unique, and reduction modulo h gives a lift-independent ordinary functional

$$D_3 : HH_0(\mathcal{C}; M_R) \longrightarrow \mathbb{Q}.$$

Positive-order corrections in the cup, cap, and basis change do not alter the cubic coefficient.

The four mixed-index Burau sign variants

$$-53824, \quad -53760, \quad -59008, \quad -59072$$

are nonzero, but they are not the selected cubic. The geometry-bound public receipt and Lean record the truncated Burau scalar

$$D_3([v_T]) = 2624 \tag{32}$$

after the endpoint-indexing erratum. The complete two-handle cocone below identifies this value with the selected MWW class, and the three-handle surface movies prove that it descends through S.

10.4. **The complete two-handle cocone.**

Lemma 10.2 (All-cable constant term)**.** *For every finite cable level and every sign-preserving cable braid b, the endpoint operator has constant term equal to its signed permutation. After standardizing that permutation, the remaining pure-braid operator is $I + O(h)$.*

Proof. BPW's oriented-cabling action is obtained from the strict BHPW tangle functor. At $q = 1$, the braiding on the fundamental representation and its dual is the ordinary symmetric interchange, including the pivotal identifications for oppositely oriented factors. Therefore the constant term of a braid is exactly the map of its signed endpoint permutation. A pure sign-preserving braid has identity permutation and hence constant term I. All matrix entries are Laurent polynomials in q, regular at $q = 1$; after $q = 1 + h$, subtracting the constant term leaves a multiple of h. The argument is independent of the number of cable copies and so holds at every finite MWW level. This statement concerns the endpoint representation only and does not assert the unknown symmetric-group factorization of the anti-parallel MWW beta action. The lemma is applied to the Johnson replacement after Lemma 10.1. ∎

Only D_3 is asserted to descend after C1. The remainder of this subsection is the C2 argument after Lemma 10.1.

P0c supplies product normals on the reconstruction strands, hence disjoint parallel levels in the replacement collar. We first type the construction at every MWW cable state. In owner order $(m_2, m_3, r_{xy}, r_{yz}, r_{zx})$, put

$$n_y = (42, 189, 2, 2, 0), \qquad n_z = (269, 1271, 2, 2, 0).$$

For $r \in \mathbb{N}^5$, cutting the actual cable $K(r, r)$ along the y- and z-cocores gives

$$p_y(r) = \sum_i r_i n_{y,i}, \qquad p_z(r) = \sum_i r_i n_{z,i}. \tag{24}$$

By Lemma 10.1, parallel copies of the P0 rectangles give a chain isomorphism H_r of the same form as (17), with the corresponding endpoint sets. The r_{zx} owner is closed by the same local counit as every other factor; an empty free word is not treated as a split unknot. Since all copies lie in one fixed tubular product chart, H_r commutes with every gluing action and with adding a parallel pair.

Let $s_0 = e_{m_2} + e_{r_{xy}}$. If $r \geq s_0$, let Ω_r be the finite set of choices of one negative and one positive physical copy of each of m_2 and r_{xy}. For $\omega \in \Omega_r$, close every unselected product factor by the MWW core counit, use the selected 44 passages for the point-push, and write $D_{r,\omega}$ for the cubic coefficient of the resulting row. Define

$$D_r = |\Omega_r|^{-1} \sum_{\omega \in \Omega_r} D_{r,\omega}. \tag{25}$$

For a state not above s_0, add the missing pairs with the once-dotted MWW maps and define D_r by pulling (25) back. Disjoint supports make the order of these additions immaterial. Thus D_r is defined on every summand in MWW Definition 3.1, including the states below the selected one.

For beta, apply the P0 collar motion simultaneously to all physical copies. This is the oriented-cabling action of [4, Theorem E], made strict by BHPW. Constant permutations reindex Ω_r. After a placement is standardized, the remaining pure braid is $I + O(h)$ by Lemma 10.2. Simultaneous transport of the operator, cup and cap, together with $\rho(W) - I = O(h^3)$, changes the row only in order at least four. Therefore

$$D_r \beta_i(b) = D_r \tag{26}$$

for every $b \in B_{r_i, r_i}$. This cubic-order argument does not use the assertion that anti-parallel braid actions factor through a symmetric group, which MWW leave open [17, Remark 3.3].

For a pair addition, split the target placements according to whether the new pair is selected. A beta move exchanges a selected new pair with an old one, so (26) reduces both subsets to the same local core calculation. On the whole source that calculation is

$$(\epsilon \otimes \epsilon)\Delta(1) = 0, \qquad (\epsilon \otimes \epsilon)\Delta(X) = 1.$$

The same local counit equations are used for every owner, including r_{zx}; they are not inferred from vanishing of a free word. Forgetting the distinguished new pair has constant-size fibers between placement sets, and the normalizations in (25) cancel their cardinalities. The dotted raw degree $+2$ is cancelled by the target cabled shift -2. Consequently, for every owner and state,

$$D_{r+e_i}\psi_i^{[0]} = 0, \qquad D_{r+e_i}\psi_i^{[1]} = D_r. \tag{27}$$

Below s_0 the second equality is the definition; the Frobenius identities and disjoint-support interchange prove the first equality and the commutation of all owner squares.

Equations (26)–(27), together with the coefficient trace relation already handled by (17), prove on the whole cabled direct sum

$$D_3(\beta_i(b)x - x) = 0, \quad D_3(\psi_i^{[0]}x) = 0, \quad D_3(\psi_i^{[1]}x - x) = 0.$$

The quotient universal property and MWW Theorem 3.2 therefore give $\overline{D}_3 : \mathcal{S}_0^2(W_2;\mathbb{Q}) \to \mathbb{Q}$.

10.5. **C conclusion and the degree-494 square.** The MWW one- and two-handle maps fit into the commuting diagram

$$\begin{array}{ccccccc} M_R(T_1,T_1) & \xrightarrow{H_{T_1,T_1}} & \mathrm{End}(U)\{-44\}\otimes A^{\otimes 227} & \xrightarrow{\mathrm{id}\otimes\epsilon^{\otimes 227}} & \mathrm{End}(U) & \xrightarrow{d_3} & \mathbb{Q} \\ \downarrow \mathrm{Glue}_{1h} & & & & & & \| \\ \mathcal{S}_0^2(W_1;K(s_0,s_0))\{315\} & \xrightarrow{\iota_{s_0}\{-4\}} & \mathcal{S}_0^{2,0}(W_1;K)/(\beta,\psi) & \underset{\cong}{\xrightarrow{\Phi}} & \mathcal{S}_0^2(W_2) & \xrightarrow{\overline{D}_3} & \mathbb{Q}. \end{array} \tag{28}$$

The left vertical map and its shift are MWW Theorem 4.7, and the bottom isomorphism is MWW Theorem 3.2. The upper selected element is $\mathrm{Id}_U \otimes X^{\otimes 227}$. Its successive degrees are

$$-44 + 227 = 183, \qquad 183 + 315 = 498, \qquad 498 - 4 = 494.$$

Diagram (28) sends it to a class x_2 satisfying $\overline{D}_3(x_2) = 2624 \neq 0$ on the Johnson replacement collar.

Theorem 10.3 (Coefficient comparison C)**.** *Hypothesis 3.3 holds for the explicit Johnson presentation.*

Proof. The 44 actual y/z rectangles and 227 actual leftover-circle transports give the coefficient-bimodule identification in Lemma 10.1; Lemma 10.2 and the Reynolds cocone verify both actions and every finite cable state. □

C/S firewall. The B-external naturality obtained in C concerns boundary cobordisms in the coefficient comparison. It does not identify the local action of an essential sphere in ∂W_2 with an ordinary dotted-sphere evaluation; that is the separate calculation in Lemma 11.3.

11. Three-handle sphere closure

This section records the intended S argument. Closed-manifold HJ Theorem 5.3 is not used to conclude that B is fixed. Remove the final 4-handle and call the remaining handlebody W_3. Its boundary is S^3. Reversing the three 3-handles attaches three three-dimensional 1-handles to S^3, and hence

$$\partial W_2 \cong \#^3(S^1 \times S^2).$$

The actual attaching sphere system has connected complement, since sphere surgery on it gives the connected boundary of W_3.

Lemma 11.1 (Kernel invariance under changing a complete sphere system)**.** *Let $\mathcal{S}$ and $\mathcal{A}$ be complete sphere systems in $\#^3(S^1 \times S^2)$. If they are related by isotopy, permutation and sphere slides, then the kernels of the two total MWW three-handle maps out of $\mathcal{S}_0^2(W_2)$ agree.*

Proof. An isotopy of an attaching sphere gives an isomorphic handle attachment. A permutation merely reorders disjoint three-handles. A sphere slide is exactly a 3–3 handle slide. Standard handle calculus gives a diffeomorphism between the resulting three-handle cobordisms which is the identity on their incoming boundary W_2. Naturality of the skein-lasagna maps therefore gives $q_{\mathcal{A}} = \Phi q_{\mathcal{S}}$ for an isomorphism Φ of the target modules. Hence their kernels agree. □

Choose a standard complete system $\mathcal{A} = (A_1, A_2, A_3)$ in the Johnson replacement reversed 1-handle picture, already disjoint from the P0 reconstruction cube B. Closed-manifold HJ Theorem 5.3 is not used to conclude that B is fixed. It is used only with Lemma 11.1: any other complete system, including a going-up Kirby attaching system, is related by isotopy, permutation and slides in the closed manifold, so the kernels agree. The current Horvat–Jabłonowski preprint [11], dated 24 August 2026, contains Lemma 5.5 (the spotted-ball slide lemma) and Lemma 5.7 (relative uniqueness of complete sphere systems). Neither lemma is invoked here: kernel invariance uses only their Theorem 5.3 together with Lemma 11.1. Lemma 5.7 concerns a relative sphere system in a manifold with spherical boundary, which is not the setting used in this section. In the reversed 1-handle picture, three belt spheres miss the P0 cube, with dual loops pairing to the identity and disjointness from the C1 leftover link and C2 supports. The $b = 0$ foams of that abstract belt model are not the MWW surfaces before the actual two-handle core disks are restored. The recorded sphere columns give the algebraic lower counts

$$(b_1, b_2, b_3) = (9920, 1430, 311).$$

Tracking the oriented dual-disk words without cancelling nonadjacent opposite copies gives the actual geometric counts

$$(b_1, b_2, b_3) = (12578, 1824, 409).$$

Their product-normal copy profiles and the corresponding $\epsilon^{\otimes b_j}\Delta^{b_j-1}$ calculations are explicit. The relative PL boundary map is represented hierarchically: it applies all 93 ambient dual-disk factors and then all six and 1513 bands of the two Kirby boundary maps, with every resulting boundary copy bound to an actual owner. The matrix itself is no longer an external input: transporting the three square compression disks of the dual handlebody through all 93 ambient Johnson factors gives $\wedge^2 A = A^{-T}$, factor by factor, and the mapping-torus boundary is $I - A^{-T}$. The separate surface-transport verifier replays the two Kirby collars and rejects a missing band.

MWW's intrinsic reformulation of a 3-handle quotient uses the local module action of $\mathcal{S}_0^2(S^2 \times D^2)$. At $N = 2$, let A_0 denote the essential sphere with one dot and A_1 the undotted sphere. The quotient relations are

$$A_0 = 1, \qquad A_1 = 0.$$

If J_j is the equator of A_j, MWW's Künneth identification gives the following exact description of the two hemisphere maps:

$$\begin{array}{ccc} \mathcal{S}_0^2(W_2; J_j) & \xrightarrow{(\Psi_{\Delta^+}, \Psi_{\Delta^-})} & \mathcal{S}_0^2(W_2) \oplus \mathcal{S}_0^2(W_2) \\ \downarrow\wr & & \downarrow \ell \oplus \ell \\ \mathcal{S}_0^2(W_2) \otimes \mathbb{Q}\{1, X\} & \xrightarrow{(\ell\circ(1\otimes\epsilon),\, \ell\circ\mu_{A_j})} & \mathbb{Q} \oplus \mathbb{Q}. \end{array} \tag{30}$$

Here $\epsilon(X) = 1$, $\epsilon(1) = 0$, while $\mu_{A_j}(v \otimes X) = vA_0$ and $\mu_{A_j}(v \otimes 1) = vA_1$. Therefore S is equivalent to the two whole-source identities

$$\ell(vA_0) = \ell(v), \qquad \ell(vA_1) = 0 \quad \text{for every } v \text{ and } j = 1, 2, 3. \tag{31}$$

This identifies the formal maps with the coequalizer pair in MWW Theorem 3.7, but not their endpoint evaluations for the actual essential spheres.

Proposition 11.2 (Fixed-detector sphere replacement)**.** *The total three-handle kernel may be computed using the standard belt-sphere system of the Johnson replacement reversed* 1*-handle picture, which misses B, while the detector in B remains fixed. For this system, S reduces to the six equations (31).*

Proof. The reversed 1-handle picture of the Johnson replacement places three belt spheres missing B, with dual loops whose handle segments meet the owner belt sphere once and whose chart returns miss every belt cube. Closed-manifold HJ Theorem 5.3 relates any other complete system to this one by isotopy in the closed manifold; Lemma 11.1 identifies the kernels. The detector remains in B, which those belt spheres miss. The six equations (31) are then Lemma 11.3. □

11.1. **The essential-sphere endpoint comparison.**

Lemma 11.3 (Endpoint factorization for a standard sphere)**.** *For every j, every cable state and every source element v, the constant term of the actual endpoint image of $\mu_{A_j}(v \otimes a)$ is the old detector image of v tensored with* *the rank-two TQFT map of the punctured standard sphere. Consequently the divided cubic row satisfies*

$$\ell(vA_0) = \ell(v), \qquad \ell(vA_1) = 0.$$

Proof. Make A_j transverse to the two-handle cocores, and let b_j be the number of core disks. MWW's proof of Theorem 3.10 represents the Δ^- hemisphere, before the core disks are restored, by the connected genus-zero surface obtained from A_j after deleting those disks and the Δ^+ disk. Choose a generic movie after cutting the one-handles. Because A_j and the detector sheet are disjoint, every critical point of the movie belongs entirely to one of the two surfaces. Mixed events are therefore only invertible Reidemeister/pivotal moves, represented at the endpoint by cable braids; there are no mixed births, saddles or deaths.

At $q = 1$, Lemma 10.2 identifies every mixed braid with the symmetry of the tensor category. Naturality of births, saddles and deaths with respect to that symmetry moves all mixed braids to the ends of the movie. The same endpoint permutation and pivotal chart transport the detector row and the surface map, so they cancel by simultaneous conjugation. Thus the actual constant map is

$$\mathrm{Id}_{\mathrm{old}} \otimes \Delta^{b_j - 1} \quad (b_j > 0),$$

and is $\mathrm{Id}_{\mathrm{old}} \otimes \epsilon$ when $b_j = 0$. The restored two-handle core disks give the b_j actual counits. Hence

$$\epsilon^{\otimes b_j} \Delta^{b_j - 1}(1) = 0, \qquad \epsilon^{\otimes b_j} \Delta^{b_j - 1}(X) = 1 \tag{32}$$

for $b_j > 0$, with the same equations interpreted as $\epsilon(1) = 0, \epsilon(X) = 1$ when $b_j = 0$. The $1, X$ basis is normalized by the Δ^+ cap in MWW Example 3.8, and BHPW strict functoriality prevents an independent sign in the Δ^- movie.

The full mixed braid maps have the form $P(I + O(h))$. The permutation P has just been cancelled, and pre- or postcomposing a row beginning in order h^3 with $I + O(h)$ changes only order h^4 and higher. Therefore the constant calculation (32) is the candidate calculation seen by the divided cubic row. For the three nonzero values $b_j = 12578, 1824, 409$, the finite Frobenius movies have respectively $b_j - 1$ coproduct saddles and b_j counits and reproduce (32). Thus the geometric ∂W_2 binding is present; the theorem below remains at the stated claim boundary until the complete MWW map is packaged in the global interface. $\square$

The endpoint exchange square is the above iterated coproduct–counit model of Lemma 11.3, bound to the actual hierarchical W_2 surfaces.

$$\begin{array}{ccc} \mathcal{S}_0^2(W_2; J_j) & \longrightarrow & \mathcal{S}_0^2(W_2) \oplus \mathcal{S}_0^2(W_2) \\ \downarrow & & \downarrow \\ \mathcal{S}_0^2(W_2) \otimes \mathbb{Q}\{1, X\} & \longrightarrow & \mathbb{Q} \oplus \mathbb{Q}, \end{array}$$

whose lower maps are the actual counit row and actual A_j-action by the movie decomposition in Lemma 11.3.

Theorem 11.4 (Relative three-handle closure S)**.** *Hypothesis 3.4 holds for the explicit Johnson presentation.*

Proof. The three H1 compression disks are transported through all 93 Johnson factors and all 1519 Kirby bands. Their geometric boundary words have lengths $12578, 1824, 409$; the resulting punctured-sphere movies realize the maps of Lemma 11.3 on the actual ∂W_2. □

12. Final identifications

MWW Theorem 3.10 identifies the iterated quotient with the module after the three-handles, and Proposition 3.4 gives the four-handle isomorphism [17]. Theorem 9.6, Theorem 10.3, and Theorem 11.4 discharge Hypotheses 3.2–3.4. For the Johnson replacement four-handle picture X_J, three-handle attachment along each belt sphere restores the boundary S^3, a PL 4-ball is attached, and the quantum degree-494 summand of the standard sphere module vanishes by the computation in the proof of Hypothesis 3.5 together with MWW Corollary 3.5. The actual E13 handle chain identifies X_J with Σ^0_A.

Remark 12.1 (Formalization boundary)*.* A Lean development [7] proves Theorem 3.1 as a conditional implication from structures `ExternalGeometry` and `CSExternalGeometry`. It checks the matrix arithmetic, the geometry-bound Burau value 2624, the degree 494, and the abstract quotient lemmas, but does not construct geometric instances. Appendix H lists the correspondence with the Lean fields.

Remark 12.2 (Earlier assumption register)*.* An earlier project draft grouped unresolved candidate geometry into five assumptions A1–A5 (actual presentation, coefficient comparison, all-state cocone, sphere maps, and rational closure). The present interface is Hypotheses 3.2–3.5 together with the uninhabited Lean fields `ExternalGeometry` and `CSExternalGeometry`. None of A1–A5 is used as a premise of Theorem 3.6. For the Johnson replacement, A1 is discharged by P0 and P3/E13, A2–A3 by C, A4 by S, and the computed part of A5 by P3/E12 together with the cited MWW Proposition 3.4 and Corollary 3.5.

13. Related work

The Cappell–Shaneson literature contains both the original construction [5, 1] and substantial standardness results [2, 9, 14, 12]. Our use of Iwaki is limited to the standard matrix form, the representative table, and the homotopy-sphere criterion. His table entry for $(41, 189, 73)$ is motivation, not a singularity or exoticness theorem.

Manolescu–Neithalath and Manolescu–Walker–Wedrich provide the relevant handle calculus for skein lasagna modules [16, 17]. BPW quantum trace functoriality and the strict BHPW tangle theory provide general comparison tools [4, 3]. Section 7 applies these functors to the actual replacement coefficient and proves the statewise naturality.

Ren and Willis use Khovanov-theoretic skein lasagna methods to distinguish other exotic four-manifolds [19]. Their computations are independent of the trace-73 construction and are not used as its comparison theorem.

Horvat–Jabłonowski's recent work gives a useful geometric criterion for recognizing three-handle attaching systems [11]. Theorem 5.3 is used only with Lemma 11.1 for kernel invariance in the closed manifold, not to fix B. The current version of that preprint, dated 24 August 2026, contains Lemma 5.5 and Lemma 5.7; neither is invoked.

14. Conclusion

For the Johnson replacement, the mathematical certificates discharge Hypotheses 3.2–3.5 and identify $X_J \cong \Sigma^0_A$. The Lean development still proves only the conditional implication because the analytic MWW objects and maps have not been constructed as an inhabitant of `ExternalGeometry`. Thus the counterexample theorem is a paper-level mathematical claim subject to expert review, not yet a fully Lean-verified theorem. Independent replay commands are listed in Appendix G.

Appendix A. Retired compact and Nielsen routes

This appendix records routes that were tested and rejected before the Johnson replacement of Section 9 was adopted. None of the material here is used in the main argument.

The symbolic AR formulas and a standard punctured-disk braid do not, by themselves, determine an embedded 44-passage collar in the reduced handlebody boundary. The explicit 19-step Nielsen representative has reduced m_2 length 309 and 42 detector channels, whereas the compact representative has length 311 and 44 channels; it is therefore not the source of the public collar word.

An inner conjugation by x^{-1} restores 44 based passages and passes a free-basis test in F_3, but leaves the unbased cyclic spine classes unchanged; it is not an embedded collar witness. A bounded search for non-inner IA corrections finds detector-changing 44-channel candidates but no signed-permutation dual-meridian extension satisfying the complementary handlebody test.

The compact straight spine words are injective but not surjective on F_3, so they cannot be the images of a handlebody-preserving $\psi_A \in \mathrm{Aut}(F_3)$. The Johnson α_{ij} factorization in the main text avoids this obstruction.

At the word level, the 309- and 311-letter representatives collect to the same normal form $y^{40}z^{269}$, but the required framing transport depends on $\mathrm{lk}(m_2, r_{yz})$, which is not determined by words alone. The railroad reduced PD constructed for the Johnson CS handle picture (Lemma 9.2) supplies $\mathrm{lk}(m_2, r_{yz}) = 0$.

Appendix B. P0 reconstruction protocol

An admissible P0 certificate consists of explicit rational PL data for a triangulated three-ball B, forty-four height-monotone labelled strands $c_i : [0,1] \to B$ with normal vectors, passage binding to the reduced attaching link, two local cancellation movies preserving framing, and an ordered crossing movie. The verifier checks passage binding, disjointness, framing transport, and letter-by-letter equality of the induced relative-endpoint braid with the public 11340-letter word.

Lemma B.1 (Soundness of the P0 certificate)**.** *Suppose an input passes the strict verifier, and suppose its independent embeddedness, disjointness and local-movie receipts are valid for the stated PL complex. Then B is an embedded framed three-ball in the reduced AR handlebody, its* 44 *strands form a framed collar, and their relative-endpoint geometric braid equals the public* 11340*-letter pure braid.*

Proof. The PL ball and the passage-binding map place every c_i in the actual AR boundary chart and preserve ownership and endpoint order through the two cancellations. Height monotonicity makes the c_i a braid in the collar; the disjointness receipt makes the strands pairwise disjoint. The normal vectors and their transport receipts give the framing on each strand and show that the two cancellation movies preserve it. Reading the certified crossing events in increasing height gives the relative-endpoint Artin word of this embedded tangle. The verifier's final equality identifies that word letter-for-letter with the public word. Since relative-endpoint isotopy classes of labelled braids are represented by their Artin words, the claimed braid equality follows. □

Remark B.2. Symbolic witnesses that import crossing rows without embedded geometry fail this contract. The Johnson certificate used in Section 9 satisfies it. The machine-checkable replay is listed in Appendix G.

Appendix C. The complete degree-three endpoint vector

With the conventions of Section 6, the coefficient vector $[\varepsilon^3](\rho_t(B_{88})-I)u = (a_0,\dots,a_{87})^T$ is

$$\begin{aligned}(a_0,\dots,a_{87}) = (&-7052,-7052,164,164,332,332,324,324,316,316,308,308,\\ &300,300,292,292,284,284,276,276,268,268,260,260,\\ &252,252,244,244,236,236,228,228,220,220,212,212,\\ &204,204,196,196,188,188,180,180,172,172,164,164,\\ &156,156,148,148,140,140,132,132,124,124,116,116,\\ &108,108,100,100,92,92,84,84,76,76,68,68,\\ &60,60,52,52,44,44,36,36,28,28,20,20,\\ &12,12,-164,-164).\end{aligned} \tag{33}$$

The detector covector is $\ell = e_{87}^* - e_2^*$, so $\ell(a_0, \ldots, a_{87})^T = a_{87} - a_2 = -164 - 164 = -328$. The sum of the 88 entries is zero, as expected for the unreduced Burau difference on the invariant total-coordinate functional.

Appendix D. The four grading contributions

term	source	calculation
-44	remove the $p(N-1)$ Hom normalization for $p = 44$, $N = 2$	$-44(2-1) = -44$
$+227$	227 distinct Frobenius labels, each of degree $+1$	$227 \cdot 1 = 227$
$+315$	one-handle gluing shift from 44 y-pairs and $271 = 44 + 227$ z-pairs	$44 + 271 = 315$
-4	cabled shift for $N = 2$, $\alpha = 0$, $\lvert r\rvert = 2$	$(1-2)(2\cdot 2+0) = -4$
total		$-44+227+315-4 = 494$

Appendix E. Convention-legality criteria

A serialization is legal only if it describes the same framed handle presentation, the same oriented based cut tangle, the same labelled boundary pairing, the same normal field, and the same insertion point. Its event order must be a linear extension of the geometric partial order. Adjacent events may be exchanged only when their supports are disjoint and a strict interchange theorem applies. Word replacement requires equality in the based labelled braid or tangle category; equality of closures, permutations, free-group words, or homology classes is insufficient.

A coordinate change P must act simultaneously:

$$W' = PWP^{-1}, \qquad u' = Pu, \qquad \ell' = \ell P^{-1}, \tag{34}$$

with the shadow, collar, basepoint, and insertion point transported in the same naturality square. Under (34), $\ell'(W' - I)u' = \ell(W - I)u$. Thus coordinate invariance is proved when the simultaneous transport exists; it is not imposed by selecting the preferred numerical answer.

A collar is legal only if it is an actual framed tubular collar of the fixed component and is isotopic to the registered collar relative to the boundary, basepoints, labels, old input balls, normal framing, and insertion point. These criteria were written before the detailed alternative collar outputs were opened. Their use in the Johnson replacement is discharged by P0.

Appendix F. Terminology

Actual map: A map induced by a specified geometric tangle or surface cobordism, as opposed to an abstract linear map having the desired source and target.

Candidate-specific binding: An identification between immutable finite data and the actual geometric object required by a published theorem.

Complete quotient: The quotient by all relation generators in the applicable MWW handle formula, not by a sampled finite subset.

Divided cubic: The reduction modulo h of a map known on its whole domain to be divisible by h^3, as in (18).

Endpoint shadow: The image of a coefficient trace class under the quantum vertical-to-horizontal trace and the fixed endpoint functor.

Finite certificate: Immutable finite data plus a deterministic exact checker. The term has no geometric force beyond the predicates actually recomputed.

Registered movie order: The oriented point-push chronology fixed in the public primitive input.

Johnson replacement: The splitting-preserving α_{ij} lift, 93-bit side choice, and CS handle picture discharged by P0–P3/E13 in this paper.

Appendix G. Extended reproduction commands

The public repository is

`https://github.com/toffee-desuwa/smooth4pc-t73-lean`.

The commands below assume a fresh clone and Python 3.10 or later.

G.1. Reconstruct the detector.

```
python -I -B scripts/recompute_t73_delta3.py --check
```

Expected bearing lines include `B44_LENGTH=11340`, `DELTA3_ETA_T1=2624`, `DELTA3_XI=0`, and `VERIFY=PASS`.

G.2. Replay Johnson P0, C, S, and P3 certificates.

```
python3 scripts/certify_t73_p0_johnson.py --check
python3 scripts/certify_t73_c1_cut_link.py --check
python3 scripts/certify_t73_c2_comparison.py --check
python3 scripts/certify_t73_s_relative_moves.py --check
python3 scripts/certify_t73_p3_four_handle.py --check
python -B scripts/certify_t73_e12_s4.py --check
python -B scripts/certify_t73_e13_close.py --check
python3 scripts/audit_t73_premises.py --check
python -B scripts/check_t73_claim_boundary.py
```

G.3. Compile and inspect the Lean development.

After installing `elan`, run

```
lake update
lake exe cache get
python -B tests/test_t73_minimal_formalization.py -v
lake lean T73Audit.lean
```

The Python test compiles the audited modules into a fresh temporary output root, requires exactly 38 axiom reports, rejects any axiom outside `propext`, `Classical.choice`, and `Quot.sound`, and rejects `sorryAx`.

G.4. **Build this paper.** From `paper/spc4-t73-candidate`:

```
pdflatex -interaction=nonstopmode -halt-on-error main.tex
bibtex main
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

The reviewed PDF is copied to `output/pdf/spc4-t73-candidate.pdf`.

Appendix H. Correspondence with the Lean fields

For readers checking the formal boundary, the principal allocation is:

Lean field	Mathematical obligation
`x0`, `ell0`, `ell0_x0`	P1 comparison and the selected genuine MWW class.
`q01`, `ell1_comp_q01`	Complete two-handle beta/psi quotient and descent.
`q12`, `ell2_comp_q12`	Reversed belt spheres of S, MWW hemisphere maps, and complete coequalizer. Not the E7 "B is fixed" claim.
`transport`, `fourIso`	E11 for the Johnson replacement X_J, not Σ^0_A.
`s4DegreeZero`	Computed `EmptyKhQ`(494)=0 and PL S^4 Euler data from `certify_t73_e12_s4.py`. The isomorphisms $G(B^4) \simeq G(S^4)$ and $\mathcal{S}_0^2(S^4) \cong \mathbb{Z}$ are MWW Proposition 3.4 and Corollary 3.5, cited not formalized. Empty-link Lean inhabitant only.
`diffeomorphismEquiv`	Graded diffeomorphism invariance of lasagna modules.
`matrixConditionsToHomotopySphere`	E13 constructed CS handle picture; Lean `CSTopologyData` uninhabited.

The geometric inputs recorded in Sections 9–12 are discharged in the paper. They have not been assembled into a single Lean inhabitant of `ExternalGeometry`; that is a formalization gap, not a hypothesis in the mathematical theorem. The empty-link control in `T73S4Inhabitant.lean` packages only `EmptyKhQ`. The Lean theorem consumes no stronger statement, and it records the external topology as structure fields rather than formalizing it internally.

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